Repository audit: diophantine-z-q Version 4

Nov 6, 2026 · @Elliot

Audit of github.com/elliotglazer/diophantine-z-q at the Version 4 state (6 Nov 2026), continuing the restricted-converse audit.

Scope and integrity

Audited commit: `fef5fb2d7fc73798aeaca1017a3b3453253e1380` (6 Nov 2026 07:47 PST, "Record verified Version 4 handoff artifacts"), 1,238 files, 80 MB, cloned read-only. The repository claims a fixed quartic P with integer coefficients in 4,973 variables such that q ∈ Z iff P(q, w) = 0 for some w ∈ Q^{4972}, with the arithmetic argument in `quartic_4973.tex` (719 KB) and finite certificate checks in `quartics/quartic-4973/`. The README states plainly that the certificate checks do not prove the arithmetic characterization and that the Lean formalization is incomplete.

Check	Result
SHA256SUMS, 15 entries	10 in-repo files match; the 5 unmatched entries are release-asset zips not stored in the tree
`polynomial.expanded.txt` and its `.gz`	both hash to `094f6a7e…`, as the README and `evidence/final-verification.json` state
Three manuscript versions (12,504; 9,625; 4,973 variables)	TeX and PDF hashes match SHA256SUMS
Earlier huge construction (`construction/`, 2↑↑7 degree)	retained for comparison; not re-run in this pass

What this audit covers: the Version 4 polynomial and its certificate, the correspondence between the pinned source system and the manuscript's definition, changes to the manuscript since the v3 text audited on 4 Nov, and the Lean status. It does not re-run the three-day source pipeline (`reproducibility/run.py`), which needs 8 GiB and a frozen runtime.

Polynomial statistics, recomputed

Every statistic in the README reproduces from `polynomial.expanded.txt` with a 40-line Python parser, and the polynomial is exactly the sum of the squares of the 4,552 packaged residuals (all weights 1), so P(q, w) = 0 over Q is equivalent to the 4,552 residual equations.

Quantity	README	Recomputed
Variables (input = index 0)	4,973	4,973 (indices 0–4972, none unused)
Total degree	4	4; degree distribution 1 / 513 / 6,364 / 12,789 / 53,258 for degrees 0–4
Nonzero monomials	72,925	72,925 distinct, no duplicates, no zero coefficients
Max absolute coefficient	685,783,949,153,203,534,233,890	same; 80 bits, below 2^80
Residuals squared	4,552	4,552 (10 linear, 4,542 quadratic); Σ r_i^2 equals the table monomial by monomial
SHA-256 of the table	094f6a7e…	same for the plain and gzip copies

The input q enters only through residual 22, $x_{57} = q^2 + 2$ (monomials $4q^2$, q^4, $-2q^2x_{57}$). So the construction defines the integers by applying an integrality predicate to $q^2 + 2$, which is legitimate since $q \in \mathbb{Z}$ iff $q^2 + 2 \in \mathbb{Z}$ for rational q. Residual 0 pins $x_{4118} = 3^{16} = 43{,}046{,}721$.

Scripts: `/home/claude/audit/repo/poly_stats.py`, `/home/claude/audit/repo/sos_check.py` (0.3 s).

Certificate checks

I ran the packaged checker on a writable copy of the package (`python3 -I verify_certificate.py .`). Result: `PASS` in 1.2 s, with a report identical to `evidence/final-verification.json`: 4,972 witnesses, 4,552 rows, 72,925 monomials, degree 4, the same maximum coefficient, and 551 operations (1 parity-conic, 13 circuit, 266

linear, 121 definitional eliminations, 2 shears, 148 eliminations modulo retained equations). The checker is a single self-contained file with the SHA-256 the manuscript prints (`b7180b01…`), and it imports nothing from the pipeline.

What a PASS establishes, reading the checker rather than its docstring:

- The pinned source `target2-tangent-first-norm-normalized-linear-pole-k40-q12-a25-h19-s32-4a82d9a8dc1b` (4,935 coordinates, 4,513 rows, degree 8, SHA `94f8a197…`) is bound by hash, and the initial quadratic lift reproduces every source row after expanding the fresh product gates (reusing a product only through a retained source row that defines it).
- Each of the 551 operations is re-derived by the checker from its own rules and compared with the supplied patch, row by row, up to a nonzero rational scalar. For the kinds actually used: `linear` is aR_i + bR_j with integer a ≠ 0; `eliminate` requires the row to be exactly v − F(…) with v absent from F; `shear` is v ↦ v + c·w; `relation_eliminate` requires the exact identity F_j(N(y), y) = G_j(y) + Σ M_jk H_k with every H_k an unchanged, x-free row; `circuit` evaluates the old and new acyclic definitions as polynomials in the external inputs and compares outputs; `parity_conic` checks the twelve-variable quadratic presentation of the seven-witness pole block against the four source residuals. Every such step has forward and reverse solution maps that fix input coordinate 0, so the rational input projection is preserved in both directions. I re-derived the parity-conic maps by hand (Section 15.9 of the manuscript); they are correct, including the b = 0 and u = −b branches.
- Occurrence checks: every removed coordinate is confirmed absent outside the replaced rows, the final rows hash to the recorded value, equal `candidate-rows.json.gz`, and after the bijective relabeling (`variable-map.json`, index 0 fixed) equal `residuals.json.gz`.
- The full integer expansion Σ r_i² is recomputed over all ordered products and compared with `polynomial.expanded.txt` monomial by monomial; degree 4 and the 2^80 bound are asserted.

What it does not establish, and says so (`scope` in the report): the source system's arithmetic characterization of the integers. Three further scope notes:

- The checker also contains rules (`unit_scale`, `redundant_inverse`, `batch_absorb`, `norm_inverse`, `localized_*`) whose soundness depends on arithmetic facts such as θ² − 2 ≠ 0 over Q. None of these is used by the Version 4 certificate; the 551 recorded operations use only the six kinds above, whose justification is polynomial identity alone.

- The checker's own source table labels this source `EXPLORATORY SOURCE; arithmetic acceptance is outside this algebraic checker` and records `ordinary_pole_soundness: negative odd exclusion; no full Psi projection`. The manuscript treats the same source as the accepted selected source. The two descriptions are consistent in substance (see the next section) but the label in the checker should be updated.
- `evidence/arithmetic-source-review.md` (6 Nov) reviews the binding of this source to the arithmetic argument at an earlier 5,173-variable endpoint and states that it "makes no claim about a later cleanup or a below-5000 result". The 4,973 endpoint uses a different certificate (the 13 circuit and 148 relation-elimination operations are new; the linear count fell from 296 to 266). Since the source is unchanged, the source review still applies; the extra cleanup is covered only by the checker, which is sufficient for the algebraic claim.

A fresh end-to-end reproduction (`reproducibility/run.py`, ~3 days, 8 GiB) was not attempted.

Source system against the manuscript

The manuscript's proof of Theorem thm:quartic (line 8690) has two layers: Theorem thm:quartic-selected-source (line 9949) asserts that the pinned degree-8 source has rational input projection exactly Z, and the certificate transports that projection to the quartic. The second layer is what the checker establishes. The first layer is the arithmetic, and it is where every nontrivial dependency of the earlier audits reappears.

Binding of the source to the text. Three checks, two of them independent of the authors' tooling:

- The source's module list in the checker (`q_anchor`, `q_primeparity`, `q_height_batch`, `q_linearnormalpole`, `q_conic13/6/10`, `q_compare4/beta/lucas`, `Z_definition`) matches the eleven modules the manuscript names, and the `parity_conic` step of the certificate checks the seven-witness pole block $\Psi(b)$ of Lemma hyp:o8-pole against four literal source rows, so that block is in the source exactly as printed (lines 7527–7534).
- In the final residual system I located the anchor: residual 0 pins x_4118 = 3^16, a chain of 21 squaring residuals carries it to x_2564 = 3^(2^25) = C, and residuals 135–136 are exactly the two equations of Lemma lem:quartic-anchor-four, $C{\cdot}H - 9C^2 - 3f_0^2 - 3f_1^2 = 0$ and $H^2 - 8U^2 - 12V^2 + 24W^2 - 4C^2 = 0$. The input enters once, as x_57 = $q^2 + 2 = \beta$.

- The authors' own source-construction auditor (`reproducibility/pipeline/check_source.py`, SHA `58bf1da4…` as printed at line 10052) re-runs here in 0.7 s on the packaged `source/` directory and reproduces `source-audit.json` byte for byte, including the construction map (`construction-map.json`). Its recorded counts agree with the text: 25 anchor squarings, 7 parity witnesses, precision 40, curve data $d = 3 - \sqrt{2}$, $D = 11 - 6\sqrt{2}$, 4,513 residuals over 4,934 witnesses. This is a replay of their auditor, not an independent reading of all 4,513 rows against Section 15; I did not do the latter.

The auditor's own report is explicit about its scope: `status: PASS: SOURCE CONSTRUCTION ONLY`, `arithmetic_acceptance: false`, `source_pin_authorized: false`, and a `not_established` list that includes "source defines the integers" and "final quartic bound". That is the right record. The README's statement that the finite checks do not prove the arithmetic characterization is accurate, and nothing in the repository claims otherwise.

What the arithmetic layer rests on. Reading Section 15 and the degree-8 Section 14 it inherits from, the source-level theorem decomposes as follows.

Ingredient	Role	Status in this audit
Five-contact height estimate, Theorem hyp:o8-height (line 6670) and its exponent-19 sharpening thm:quartic-height19 (8738)	the only bound on heights of the root points; drives both contradictions in Theorem thm:quartic-selective-arithmetic (9604)	proof read in full; arithmetic of (LR4)–(LR6), (LT1)–(LT7), the CM bound 854R < 6^100 R^24, the Swan-conductor bound 274, and the numerical certificates (B_* has exactly 109,445 bits, $\pi(76176) = 7499$, largest prime 76163, $\log_2 H <$ 4,783,227, $\log_2 H_{19} <$ 2,093,966) all re-derived and correct. External inputs at citation level: Pan 1.0.4 at $\ell = 5$, Carayol local-global compatibility, Yuan 2024 integral moduli, González–Rotger quotient models, Gaudron–Rémond 2023 Thm 1.9(1), Javanpeykar 1.1.1, von Känel 2021 (2.2) and (5.9), Faltings, Serre. The same structure as the six-contact proof verified on 4 Nov, with different geometry (discriminant-210 Shimura curve, [L:Q]

Ingredient	Role	Status in this audit
Restricted converse RC (Theorem hyp:RC, 1020)	completeness of **both** pole modules: the seven-witness Ψ block (Lemma lem:original-seven-parity-completeness, 7708, through the simultaneous-realization Lemma at 7603, which applies RC to E_n and E_{nθ}) and the linear-unit ordinary pole (Prop. lem:quartic-linear-pole, 9393, "under the same inherited realization and rank hypotheses")	unchanged from the earlier audits; the quartic's completeness direction (Z ⊆ projection) is conditional on RC exactly as the huge polynomial was. Soundness (projection ⊆ Z) does not use RC.
Pole soundness: Lemma lem:original-seven-parity-soundness (7569) and the one-sided part of lem:quartic-linear-pole	excludes negative odd valuations of the pole argument outside {2, 3, 17}	both re-derived line by line; correct, including the zero-ordinate branches and the b = 0 case
Lucas growth, Lemma lem:o8-lucas (8565)		t
The two height contradictions, Theorem thm:quartic-selective-arithmetic (9604)	from the local congruences and the height bound to Δ = 0 and then 2β ∈ Z	elementary estimates checked:
Quaternion trace anchor, Lemma lem:quartic-anchor-four (9487), citing Koenigsmann Prop. 6	encodes 2,3-integrality of f(S)/C in two quadratic equations	the local facts (algebra (2,3) ramified exactly at 2 and 3; local norm-one traces at a ramified place are exactly Z_p; the parity cases of 9 + 3(a² + b²)) re-derived; correct
Quadratic presentation of the parity block (10128)	the one `parity_conic` certificate step	forward and reverse maps verified by hand; correct

Ingredient	Role	Status in this audit
Fixed-system prime-pattern theorem with prescribed finite exclusions (Prop. prop:fixed-prime-pattern, 1282; Green–Tao–Ziegler arXiv:1009.3998v5 with the author erratum, plus Bennett–Martin–O'Bryant–Rechnitzer)	existence of the contact primes in the completeness direction	**not audited** in any pass of this session
Generator certificate $2E(K) = Z\cdot P_*$, $E(K)_{tors} = E2$ for $y^2 = x^3 - Dx$ (Prop. prop:generator, 283); formal-group logarithm and valuation lemmas; reciprocal-logarithm polynomials G_9, G_2 and their dyadic nonvanishing (9024–9116); the private-conic, joint-graph, target-two and normalized-Lucas lemmas of Section 15	the local machinery that turns witnesses into congruences $C_i \equiv n_i \bmod p^{40}$	read at statement level only; not re-derived

The upshot is unchanged from the earlier documents: the quartic is a faithful compilation of a source whose arithmetic theorem is conditional on RC (for completeness) and on the five-contact height estimate with its cited inputs (for soundness). The repository is careful to say so.

Manuscript changes since the v3 text

`quartic_4973.tex` (14,982 lines) against `main_v3.tex` audited on 4 Nov (10,692 lines): 178 lines removed, 4,468 added. The additions are almost entirely two new sections and one appendix; the core chain audited earlier (height Section 4, RC Sections 7–9, Appendices E–G) changed only in polish.

New material.

- Section 14, "The base residual compiler" (6606–8682): the degree-8 compiler F_8 (129,497 monomials, 15,671 witnesses), the five-contact height theorem hyp:o8-height with its own geometry (discriminant-210 Shimura curve, González–Rotger models,

Yuan's integral moduli, a field L with [L:Q] | 576, the bounds (R), (G), (LR1)–(LR6), (LT1)–(LT7)), the seven-witness pole predicate Ψ with soundness and RC-dependent completeness, the elliptic charts, norm tests, forty-one contacts, scalar height and the Lucas growth lemma.

- Section 15, "An explicit quartic defining the integers" (8683–10236): the quartic theorem, the exponent-19 height sharpening, the thirty-seven-contact and reciprocal-logarithm lemmas, the three-mode source (full / beta-only / Lucas-only comparisons), private conics, the linear-unit ordinary pole, the four-witness quaternion anchor, the normalized Lucas tests, the two height contradictions, the selected-source theorem, the quadratic reductions and the proof of the quartic theorem from the certificate.
- Appendix "De Rham comparison and finite descent in the height argument" (14583): Lemma lem:de-rham-descent via Scholze and Berger, supporting the finite-descent step of the height proofs.

Edits inside the audited chain (all presentational or citation-tightening; none changes a statement or an inference I had verified):

- Sutherland's division-polynomial conventions made explicit; Milne's abelian-variety references added.
- Conrad's local algebraicity cited so that the twist in Pan's theorem is finite; Li and Shimura cited for the abelian variety A_f.
- An explicit Herbrand formula G^υ ∩ G_F = G_F^{ψ(υ)} for υ > max(4D², 3e) replaces an informal upper-numbering remark.
- Lin–Yang / LNY theta-null conventions written out (j = 3779136 (Π z_m)^4 / (Σ z_m^8)^5).
- A new Lemma lem:forward-rank, "Forward analytic rank zero and one" (1101), packages BCDT plus Kolyvagin; it is what the pole-completeness lemmas call the "published forward analytic-rank-one theorem".

Status of the earlier open rows. Nothing in the new text reopens them: the e_q unit, the a = 0 pairing, the quadratic exception, the Lemma 9.2 constant, Δ³, the G.4 cross terms and the F.2 lift independence all stand as closed on 4 Nov. The six-contact height section is textually the v3 section I verified, with the citation polish above.

One presentational point: Section 14 still carries comment lines such as `% UNAPPLIED CANDIDATE: global review and published Pan match are pending.` (7243) and `% This lemma is conditional on the separately audited field/ramification lemma.`

(7350). They look like leftover pipeline annotations rather than live caveats, since the surrounding text proves the lemma in question, but a reader of the source will not know that.

Lean formalization

`lean/` is a snapshot of 48 helper modules (6,553 lines, 463 public and 69 private declarations) pinned to Lean 4.35.0 and a fixed Mathlib revision. `scripts/check_lean.py` runs here without a toolchain and reports `source_check: PASS, byte_identical_modules: 48, main_theorem_proved: false, quarantined_sorries: 1`. No Lean toolchain is available in this container, so the modules were not compiled; the README's `--build` path, which rebuilds the 48 modules and rejects any axiom beyond `propext`, `Quot.sound` and `Classical.choice`, was not exercised.

Three observations:

- The single `sorry` is the whole theorem. `Frozen/BoundedIntegerPolynomial.lean` states `exists_bounded_integer_polynomial_defining_integers`: a polynomial over Int in `Fin 292890231905` variables, total degree below 2↑↑7 and coefficients below 2↑↑8, whose rational zero set projects onto Z. That is the statement of the original huge polynomial, not of the quartic. Nothing in `lean/` mentions the 4,973-variable quartic, its degree 4, or its coefficient bound.
- The helpers that touch the arithmetic are for older machinery. `ZInQ/PoleValuation.lean` proves `poleSystem_no_negative_odd_valuation` for a pole system with six Boolean coordinates, an inverse equation e·H·z·w = 1 and multiplier (1−2b_0)(1+b_1)…(1+12b_5), excluding primes dividing 2730. That is neither the seven-witness Ψ block nor the linear-unit pole of Version 4, so the Lean pole lemma does not cover the quartic's pole soundness (which I checked by hand instead). The `Alpha*`, `Precision*`, `Circuit*`, `HeightConstant*` modules likewise belong to the huge-polynomial compiler.
- The README is accurate: "This is not a formal proof of the 11/5 manuscript's main theorem", and the height, restricted-converse, pole-completeness, multiplication, compiler and composition obligations are listed as open.

For Version 4 specifically, the natural formal targets that would add value, in increasing difficulty, are: the sum-of-squares identity and the input-projection equivalence of the certificate (purely algebraic, machine-checkable from the packaged files), the seven-

witness and linear-unit pole soundness lemmas (elementary valuation arguments of the kind already in `PoleValuation.lean`), and the Lucas growth lemma. None of these touches RC or the height estimate.

Findings

Short answer. The repository delivers what it says: an explicit degree-4 polynomial in 4,973 variables, a certificate proving it has the same rational input projection as a pinned degree-8 source, and an honest statement that the source's arithmetic theorem is the manuscript's job. Every finite claim I could recompute is correct. The arithmetic theorem is conditional in the same two places as before: completeness needs RC, soundness needs the five-contact height estimate with its cited inputs. No new gap appeared; no old gap closed.

Item	Verdict
README statistics, SHA256SUMS, table ↔ gzip, table = Σ r_i^2	verified by independent recomputation
`verify_certificate.py` PASS; soundness of the six operation kinds used; parity-conic maps	verified (checker read, maps re-derived by hand)
Source ↔ manuscript binding (anchor chain, anchor equations, Ψ block, module list; authors' source auditor replayed, output identical)	verified at the level stated; the 4,513 rows were not read against Section 15 one by one
Five-contact height proof hyp:o8-height and thm:quartic-height19: internal arithmetic and numerical certificates	verified inference, conditional on the cited external theorems (Pan, Carayol, Yuan, González–Rotger, Gaudron–Rémond 2023 1.9(1), Javanpeykar, von Känel)
Pole soundness (seven-witness and linear-unit), Lucas growth, two height contradictions, quaternion anchor, anchor size vs. (24H_*)^10	verified inference
Pole completeness, hence Z ⊆ projection	conditional on RC, as in every earlier version

Item	Verdict
Prime-pattern theorem (Prop. prop:fixed-prime-pattern; Green–Tao–Ziegler + BMOR)	not audited in this session
Section 15 local machinery (private conics, joint graph, target-two congruences, reciprocal logarithm G_9/G_2, generator certificate)	statements read, proofs not re-derived
`evidence/arithmetic-source-review.md`	covers the 5,173-variable endpoint and disclaims the cleanup to 4,973; the source is unchanged so the review's substance carries, and the cleanup is covered by the checker
Lean	48 helpers, main theorem `sorry`; the frozen target is the huge polynomial, not the quartic

Needs clarification (labels and leftovers, not mathematics).

- The checker's source table marks the accepted source `EXPLORATORY SOURCE`, and `source-audit.json` records `source_pin_authorized: false`; the manuscript and README treat the same source as accepted and pinned. Align the labels or say what "authorized" means.
- Comment lines in Section 14 (`% UNAPPLIED CANDIDATE … published Pan match are pending`, 7243; `% … conditional on the separately audited field/ramification lemma`, 7350) read as unresolved caveats; if they are pipeline residue, remove them.
- Theorem thm:quartic says "with precisely the inherited arithmetic hypotheses and qualifications" but does not name them. Since the statement is what will be quoted, it would be clearer to state in the theorem itself that the equivalence is proved assuming RC (Theorem hyp:RC) and the five-contact height estimate (Theorem hyp:o8-height), with the latter's cited inputs listed in Appendix B.
- The README's wording "The manuscript presents a fixed polynomial … such that, for every rational q, q ∈ Z iff …" is unqualified, while the Verification status paragraph two sections later carries the qualification. A one-clause pointer in the first sentence would prevent the headline from being read as unconditional.

Discrepancies found: none in any recomputed quantity, hash, count or elementary inference.

Next checks, in the order I would do them.

1. The restricted-converse chain remains the main outstanding mathematical obligation; the remaining external checks are the ones listed under Next checks in the restricted-converse document (Cornut–Jetchev, Waldspurger Cor. 2, the published form of Pan's theorem).
2. Audit the prime-pattern input (Section 6, Prop. prop:fixed-prime-pattern) against Green–Tao–Ziegler and the Bennett–Martin–O'Bryant–Rechnitzer bounds; it has not been read in any pass.
3. Read the Section 15 local lemmas (9205–9910) with the same line-by-line method used for Section 8; the two I re-derived were correct, which is mild evidence for the rest but not a check.
4. Confirm Gaudron–Rémond 2023 Theorem 1.9(1) in the published Mémoire, since Version 4 now uses it in three places (CM bound, (LT3), and the 19-exponent sharpening); the 4 Nov audit verified the 2014 Theorem 1.4, not this one.
5. Optionally, a clean-room reproduction (`reproducibility/run.py`) on a machine with 8 GiB and three days, to close the loop from the module contracts to the released polynomial.

Update: Versions 5 and 6 (6 Nov, 10:06)

What changed since commit fef5fb2d

Audited head: `f87226569fa7ceeae0c783c3e4d1241ac727733a` (6 Nov 2026 10:06 PST), fourteen commits after the morning snapshot, 655 files changed, about 13 million lines added (almost all of it JSON intermediate representations under `quartics/quartic-1272/` and `quartics/quartic-962/`). SHA256SUMS now has 28 entries; all 18 in-tree files match, and the 10 unmatched entries are release zips not stored in the tree, as before.

In order:

- `research/2026-11-06-collaborator-notes/`: notes from the project's collaborators (the README names eight contributors), a height-replacement note that reduces the notes' von Känel–Kret dependence to the existing five-contact proof, a comparison of ideas, and a copy of the morning's audit record (`fable-repository-audit.md`) with a source-status clarification answering the `EXPLORATORY SOURCE` label raised there.

- Credit to Pheidas (rank-one curves and local divisibility) added to all manuscripts; the only change to `quartic_4973.tex` since the audit is that paragraph and its bibliography entry (12 lines).
- Version 5: `quartic_1272.tex` and `quartics/quartic-1272/`, a 1,272-variable quartic reconstructed from the public agentic-improvements notes (Brox), the baseline for Version 6.
- Version 6: `quartic_962.tex` (19,034 lines) and `quartics/quartic-962/`, a 962-variable quartic obtained from the 1,272 baseline by changes of coordinates, smaller circuits and 32 certified eliminations; a new 2,450-line appendix proving the simultaneous-prime input; a new section proving that a single defining polynomial must have degree at least four; `docs/quartic-962/` inventory with 125 external application families.

The retained text is untouched: `quartic_962.tex` is `quartic_4973.tex` with the title, abstract and one introduction paragraph edited, Sections "Arithmetic interfaces for the smaller quartic" and "The shared-point quartic in 962 variables" and "Why a single defining polynomial must have degree at least four" inserted after Section 15, the appendix "Simultaneous primes with coprime adjacent norms" inserted before the bibliography, and new bibliography entries. Everything audited on 4–6 Nov (the RC chain, the six- and five-contact height proofs, the 4,973 source) is byte-identical.

The README's headline sentence now reads "three integer-coefficient polynomials whose rational zero projections it proves to be the ordinary integers", with the qualification three sections down ("These checks do not prove the source system's arithmetic characterization"). The point raised in the morning's findings about the headline stands.

Version 6 polynomial, recomputed

`quartics/quartic-962/polynomial.json` (SHA `6b798de7…`, as stated) lists 55,409 monomials whose coefficients are integer polynomials in the symbol S = 3^(2^27). I parsed it with my own script and treated S as an indeterminate throughout (`/home/claude/audit/repo962/sos962.py`, 0.9 s).

Quantity	README	Recomputed
Variables (input = index 0)	962	962, indices 0–961, none unused

Quantity	README	Recomputed
Degree	4	4; distribution 1 / 141 / 1,713 / 7,143 / 46,411 for degrees 0–4
Monomials	55,409	55,409
Coefficients independent of S / linear in S / equal to $64S^2$	55,401 / 7 / 1	same; the S^2 term is $64S^2 \cdot x_{79}^2$ and the seven S-linear terms are the cross terms of the anchor residual
Largest S-free coefficient	not stated	1,526 bits (460 digits), so $64S^2$ = 64·3^(2^28) is the maximum, and $\log_2$ of it is about 4.25×10^8 < 429,496,736 as the manuscript says
Primitive content	1	1 (gcd of all coefficients, S-powers included)
Sum of squares	919 residuals	Σr_i^2 over the 919 residuals of `compiled-ir.json`, after the compaction in `coordinate-map.json`, equals the table monomial by monomial as polynomials in S; all residuals have degree ⩽ 2 (so with S = 3^(2^27) substituted the identity holds over Z)

The input enters once, as x_412 = q^2 (source numbering): this construction applies its integrality predicate to q^2, not to $q^2 + 2$ as Version 4 does. Both are fine since $q \in Z$ iff $q^2 \in Z$ for rational q.

Version 5 (`quartics/quartic-1272/`) also reproduces: its check script reports 1,272 variables, 48,001 monomials, 1,229 residuals, the same height $64S^2$, byte-identical artifacts, in 12 s. I did not recompute its table independently; Version 6′s certificate chain starts from a hash-pinned copy of it.

Version 6 certificate and source program

Packaged checks. `python3 -E -B scripts/check_quartic962.py --out <fresh dir>` runs in 5.4 s here and reports PASS on all fourteen stages (package integrity, serialization policy, baseline source bindings, accepted equivalence evidence, compiler self-test, build,

compile-translated, independent certificate check, emitted quotient semantics, point-chart equivalence, emitted polynomial circuits, anchor equivalence, frozen artifact identity, fixed integer statistics), with the seven stable artifacts byte-identical to the frozen package. The inventory validation (`docs/quartic-962/inventory.py validate`) and its 25 adversarial binding tests also pass.

Independent replay of the reduction. The chain the package certifies is: baseline P_{1272} → (witness maps in `PROOF.md`) → raw system `raw-ir.json` (994 coordinates, 951 residuals) → anchor translation `raw-translated-ir.json` → 32 eliminations → `compiled-ir.json` (962, 919) → Σr_i^2 = `polynomial.json`. I re-verified the last three arrows myself (`/home/claude/audit/repo962/replay962.py`, 0.3 s):

- `raw-ir` and `raw-translated-ir` differ in exactly one residual, the anchor row 950. The raw row is $u^2 + 6v^2 - 2w^2 - 3A^2 + 2S^2 = 0$, which is equation (eq:v5-physical-anchor) of the manuscript; the translated row equals 16 times the raw row after the invertible affine change $u = S + (U - 3V + 3W)/4$, $v = S/2 + V/4$, $w = 3S/2 + W/4$ (I expanded it exactly: $U^2 - 6UV + 6UW + 15V^2 - 18VW + 7W^2 - 48A^2 + 8SU$; the S^2 terms cancel, and squaring $8SU$ is the origin of the single $64S^2$ coefficient).
- Each of the 32 `eliminate` operations names a pivot residual that is exactly $v - F$ with F free of v; substituting $v := F$ in every residual and dropping the pivot leaves 919 residuals of degree $\leqslant 2$ in 962 coordinates, equal to `compiled-ir.json` up to a nonzero rational scalar per residual. The remaining 862 operations are those scalings.
- The sum of squares matches the table (previous section).

So for every rational x, the raw system has a rational solution with input x iff $P_{962}(x, \cdot)$ has a rational zero. That part is now verified independently of the authors' checker.

What I did not replay. The arrow from the 1,272 baseline to the raw system is the content of `PROOF.md`: new point charts (t, h with $u = t^2$, $X = h^2$, $u(X - u/4) = D$, $Y = h(u/2 - X)$), regrouped circuits, merged quotients, rescaled pole witnesses ($z = C \cdot e$, $C = 17b$) and a signed-partial-product correction chain. I checked by hand the point-chart maps in both directions ($t = b/a$, $h = (a^2 + D)/(2b)$ recovers the doubling formulas; $a = th + t^2/2$, $b = at$, $m = h + t$ gives back $a^2 - t^2 a = D$, $b^2 = a^3 - Da$, $2bm = 3a^2 - D$), the pole rescaling (the new first residual is C times the old one, and $H = Ch$) and the correction chain ($c_0^2 = 1$, $(c_i - c_{i-1})(c_i - p_i c_{i-1}) = 0$ gives exactly the 256 signed squarefree corrections). The quotient-domain claims (every scale A positive, $F \cdot e = (A\delta)^5$ forcing $F, e \neq 0$, $W \neq 0$ because $w_0 \neq 0$) read correctly but depend on baseline lemmas I have not read. The packaged `point_chart_equivalence`, `emitted_quotient_semantics` and `anchor_equivalence` stages test exactly these identities on the emitted gates.

Metadata. `compiled-ir.json` still carries `variant.status: pending_independent_review` and the source record `reconstruction: independent_from_displayed_formulas_not_original_gate_bytes` (the gate program was rebuilt from the formulas displayed in the Brox notes, not from an original table, which the README also says is unavailable). The package README explains that such strings are historical; the Version 6 inventory review records acceptance of the final artifacts.

The new analytic input: simultaneous primes with Gaussian ray-class conditions

The README calls this "the needed Gaussian ray-field estimate". Precisely, it is Theorem thm:v5-simultaneous-primes (line 16180): for distinct integers $c_1, \dots, c_k$, any q, any odd squarefree M prime to 7, and any H, Y, there are u, w with $q \mid w$, $u/w^2 > H$, all $p_j = u + c_j w^2 > Y$ prime, each with a Gaussian generator $\pi_j \equiv 9 \pmod{16}$, $\equiv 1 + 3i \pmod 7$, $\equiv i \pmod M$, such that the "adjacent norms" $N(\pi_j \pm 1)$ have no odd common prime factor within or across positions, with $v_2 = 6$ and 2 and cross gcd exactly 4. The adjacent norms are the orders of the reductions of the curve at the two places over p_j; the theorem is what lets the completeness proof of the 962 source choose nine detectors' worth of contact primes at once (lines 11083–1131). The height input of the 962 construction is unchanged: it is the five-contact theorem with exponent 24 that I read in full this morning, not the exponent-19 sharpening.

The statement is attributed to Sun, *86 unknowns for Hilbert's tenth problem over the rationals* (arXiv:2610.20691v2, Theorem 4.1), whose proof uses Frei–Tinková's *Linear constellations in primes with arithmetic restrictions* (arXiv:2408.17441v3, listed as to appear in Mathematika). The manuscript elects not to assume Frei–Tinková and instead proves the needed special case (fixed ray class fields of $\mathbb{Q}(i)$) in a 2,450-line appendix (16159–18611). Its architecture:

1. Local counts on the quadric $Q_\kappa = a^2 + b^2 - c^2 - d^2 - \kappa w^2$ and a Selberg upper sieve for pairs of positions sharing an odd prime divisor in a medium range $L < \ell \leqslant B^{1/100}$, with the uniform congruence count taken from Browning–Loughran (TAMS 2019, Theorem 4.1).
2. A large-prime tail from the geometric sieve for quadrics of Browning–Heath-Brown (Forum Math. 2021, Lemmas 2.1 and 4.1), giving Proposition prop:v5-obstruction-tail.

3. Gaussian ray-class prime weights, their main terms in progressions via Hecke characters and Mitsui's 1956 generalized prime number theorem, a Gaussian ideal Vaughan decomposition with Type I/II estimates against nilsequences using Green–Tao's *Quantitative behaviour of polynomial orbits on nilmanifolds* (Theorem 2.9, Lemma 4.5, Appendix A) and the Green–Tao Möbius paper, and Bienvenu's higher-dimensional Siegel–Walfisz theorem; the output is the orthogonality estimate (eq:gaussian-centered-nil-input): the ray-class prime weight minus its density times Λ is o((log L)^{−B}) against any fixed nilsequence.
4. Transference: Green–Tao *Linear equations in primes* (Prop. 6.4, App. D, Prop. 7.1, App. C, Lemma 4.4) for the majorant and generalized von Neumann theorem, and Dodos–Kanellopoulos (Acta Arith. 2022, Theorem 5.1) as the relative inverse theorem, with the stated exponent bookkeeping (target U^{u+1} needs majorant norm U^{2u+2}).
5. Tao–Ziegler, *Polynomial patterns in the primes* (Forum Math. Pi 2018, Theorems 9, 11, 13, Cor. 3) for the quadratic pattern u + c_j w², combined with the Artin weights into Proposition prop:v5-artin-patterns, and a final diagonal choice of parameters (L first, then the ray field, then τ_j ↓ 0 and N_j) in subsec:v5-diagonal.

What I checked: Lemma lem:v5-local-counts in full (the ℓ^4 point count, the $4\ell^2$ bound and the inert-case uniqueness for G_ε = H_η = 0, the exact density g(ℓ) and its expansion (k + 2χ)/ℓ + O(ℓ^{−2}), and the k = 2, χ = −1 value 2/(ℓ² + 1)) — correct; the proof of the theorem from its propositions (order of limits, the 2-adic valuations from π ≡ 9 mod 16, the removal of small odd divisors by the ray conditions, u/w² → ∞ from w ⩽ τ√N) — correct as written; and that the Tao–Ziegler hypothesis is met (the degree-2 coefficients c_j are distinct and the system u + c_j w² is admissible). The theorem is in any case consistent with what Tao–Ziegler already give without the Gaussian refinement: polynomial patterns x + P_j(m) with P_j(0) = 0 inside any set of primes of positive relative density, with m ⩽ log^L x, which yields the primes, the ray-class conditions and u/w² → ∞ but not the pairwise coprimality of adjacent norms; the latter is what the sieve in items 1–2 is for, and it needs a count rather than infinitude, which is why item 3–5 reprove an asymptotic.

What I did not check: items 2–5 at the level of inference. That is roughly 2,000 lines of Green–Tao–style analysis (Type I/II ranges and exponents, the concatenation of local Gowers norms, the W-trick with a cutoff that varies with the physical scale, the Hecke-character main terms). It is the single largest piece of new mathematics in the repository, it is entirely AI-written, and the only reviews of it in the tree are internal model reviews (docs/quartic-962/evidence: the Type I/II root review, the norm-transfer, factorization and Siegel–Walfisz independent reviews, and the end-to-end integration review that accepts the removal of the assumed Frei–Tinková theorem). Those reviews are substantive and I

had overlooked them in the first draft of this section, but they are not external mathematical validation, and the integration review itself says its Siegel–Walfisz component is its own author's work. For a conditional reading, the cleaner statement is: the 962 construction depends on Sun's Theorem 4.1, which depends on Frei–Tinková Theorem 1.4 (accepted, unpublished); the appendix is an attempt to remove the second dependence, and until it is audited the dependence should be listed as still present.

Also worth recording: with this appendix, the 962 construction no longer uses the fixed-prime-pattern theorem of Section 6 (the Green–Tao–Ziegler linear-forms route) that the 4,973 and original constructions use. The two constructions therefore rest on different prime-pattern inputs, neither of which has been audited here.

The 962-variable arithmetic argument and the degree bound

Dependency structure. Theorem thm:v6-962-polynomial (11217) reduces to Theorem thm:v5-source-projection (11004), "with the inherited arithmetic results and Theorem thm:v5-simultaneous-primes". Reading the proof, the inherited results are: the five-contact height theorem with exponent 24 (soundness), the original seven-witness pole predicate Ψ and its completeness through Lemma 7603 (hence RC), Lemma lem:o8-lucas, the generator certificate and the formal-group lemmas, plus the new simultaneous-prime theorem (completeness). The bridge lemmas of the new Section "Arithmetic interfaces" connect the old objects to the new source:

- v5:scalar32-bridge (10260): the norm tree with orders (1,1,1,1,32,32) and the pole obstruction force $M_5(\beta)^{32} \mid a$ and $h(\beta) \leqslant H_*|a|^{3/4}$. I re-derived the valuation descent through the tree ($R \geqslant 1$ from the constant leaf, $v(W) > -R/2$, then leaf bounds $-9R/16$, $-5R/16$, $-13R/16$, $-3R/8$, $-7R/16$, $-15R/16$, all $> -R$) and the formal-logarithm step $v_p(z(S)/z(J_0)) = v_p(a/h_0)$; correct.
- v5:preparation-zero-bridge (10395): the pole predicate with $t = 0$, $U = b$; soundness inherited; completeness via the simultaneous-realization lemma under $\mu \equiv 17$, $\theta \equiv 1 \pmod{24}$, disjoint supports and mutual local squareness. I re-derived those congruences from $v_2(b) \geqslant 3$, $v_3(b) \geqslant 1$, $v_{17}(b) \geqslant 2$, $b < 17$, and the explicit witness formulas $e = e_0/B$, $p = y_1/B$, $q = y_2/(B^2 s)$; correct.
- lem:v5-three-witness-anchor (10811): the new anchor $u^2 + 6v^2 - 2w^2 - 3(f_0^2 + f_0 f_1 - f_1^2)^2 + 2S^2 = 0$ is equivalent to f_0/C, $f_1/C \in Z_{(2)} \cap Z_{(3)}$ via Voight 14.6.7 for the algebra $(2,3)$. I re-derived the local analysis of $1 - 2/(3t^2)$ at 2 and 3, the anisotropy of $A^2 + AB - B^2$ over F_2 and F_3, the implied label condition $a \in 2 \cdot 3^{(2^{25} - 1)}Z$ and the size comparison ($\log_2 \min|a| > 50{,}331{,}647$ versus $\log_2(21H_*)^{13/2} < 31{,}091{,}008$); correct.

- Soundness of thm:v5-source-projection: the two height contradictions with exponents 24/52 and 24/16 (B^{2/13} < 21H_* against |a| > (21H_)^{13/2}; log B < 2log(32H_) + 3h(β) < |a|/4 against the Lucas bound log B > |a| − 3); arithmetic checked, including $5^6 \cdot 7^{13} < 15^{13}$.
- Not re-derived: lem:v5-two-anchors, lem:v5-height-transfer, lem:v5-one-place, lem:v5-unramified-log, the two reduced group cardinalities, and the detector/pole domain claims (R_r, W_r, T_r, B_r) in the completeness proof.

So, conditional on the same two inputs as before (RC and the five-contact height estimate with its cited theorems) and on the new simultaneous-prime theorem, the 962-variable quartic defines Z. Nothing in the new material weakens or strengthens the status of RC or of the height estimate.

The degree-four lower bound (Section 11561–11736, from the agentic notes): if $S_F = Z$ for a single $F \in \mathbb{Q}[T, Y_1, \ldots, Y_r]$ then $\deg F \geqslant 4$. I read the proof in full. Degree $\leqslant 2$ is excluded by a three-point argument on a line or a conic (the conic has a smooth rational point because an irreducible quadric has at most one geometric singular point; a rational parametrization then contradicts the lemma that a nonconstant rational function cannot be integer-valued on the rational points of a real ball). A reducible cubic is excluded because each factor projects to at most two integers. For an irreducible cubic, geometric reducibility (three conjugate hyperplanes) is excluded by the line through the zeros with inputs 0 and 1; the vertex subspace W is quotiented out, and the two chosen zeros force X_0 and X_1 to vanish on W so the input survives the quotient; for quotient dimension $\geqslant 2$, Kollár's theorem (cubic hypersurface, geometrically integral, not a cone, with a rational point ⇒ unirational over Q) and the ball lemma finish; for dimension 1, the integral form of Hilbert irreducibility supplies an integer t with no rational Y. Every step is correct as written. The two external inputs are Kollár 2002 Theorem 2 (the manuscript notes the journal statement was not compared with the author version) and integral Hilbert irreducibility as recalled in Müller 2002. This is an unconditional result, independent of RC and the height estimate, and a nice one: the quartics are of optimal degree.

Credits. The manuscript now credits Pheidas (2000) for the rank-one/local-divisibility strategy, Sun for the prime-pattern strategy, and the collective agentic notes (Brox's public repository at a pinned revision) for the shared-point construction, the fixed height scales, the quadratic compilation and anchor translation, and the degree obstruction. CITATION.bib and the author field remain blank by design.

Findings for Versions 5 and 6

Short answer. The finite content is correct and now verified independently end to end for Version 6: the 962-variable quartic is exactly the sum of squares of the packaged residuals, which are exactly the 32-fold elimination of the raw gate system, whose anchor translation I expanded by hand. The packaged checkers for Versions 5 and 6 pass here. The arithmetic status is: Version 6 is conditional on RC, on the five-contact height estimate with its cited theorems, and on the simultaneous-prime theorem of the new appendix, which replaces the Section 6 prime-pattern input and is the largest piece in the repository that no one outside the project's own agents has read. "Conditional" throughout this document means a dependency I have not validated, not an inference I found to fail. The degree-four lower bound is unconditional and checks out.

Item	Verdict
Version 6 table statistics, content, height, SOS identity (S symbolic)	verified by independent recomputation
32 eliminations raw → compiled, anchor translation, input fixed	verified by independent replay
Baseline 1,272 → raw 994/951 (point charts, circuits, quotient merges, pole rescaling, correction chain)	packaged identity checks pass; point charts, pole rescaling and correction chain re-derived by hand; quotient domains depend on baseline lemmas not read
Version 5 package	packaged checks pass (12 s); table not recomputed independently
Bridge lemmas (scalar32, preparation-zero, three-witness anchor) and the soundness arithmetic of thm:v5-source-projection	verified inference
Completeness of the 962 source	conditional on RC (via Ψ completeness) and on thm:v5-simultaneous-primes
thm:v5-simultaneous-primes: statement, local-count lemma, final assembly, Tao–Ziegler hypotheses	verified; consistent with Sun Thm 4.1 / Frei–Tinková Thm 1.4

Item	Verdict
thm:v5-simultaneous-primes: sieve propositions and the Gaussian nilsequence / transference argument (items 2–5, ~2,000 lines)	**not audited** by me; five internal agent reviews in docs/quartic-962/evidence, no external review
Degree-four lower bound	verified inference, conditional only on Kollár 2002 Thm 2 and integral Hilbert irreducibility
Retained text (Sections 1–15, Appendices A–G, de Rham)	byte-identical to the audited Version 4 text apart from the Pheidas paragraph
SHA256SUMS, inventory validation, 25 binding tests	pass

Needs clarification.

- The README's first sentence ("three … polynomials whose rational zero projections it proves to be the ordinary integers") and the Verification-scope paragraph are three sections apart; the first should carry the qualification.
- The README says Version 6 "proves the needed Gaussian ray-field estimate within the manuscript … without assuming their theorem", while `quartics/quartic-962/PROOF.md` and the package README still say the projection claim "retains the baseline's arithmetic qualifications, including its current Frei–Tinková input". The two should agree; until the appendix is reviewed, the package wording is the safer one.
- `compiled-ir.json` still says `pending_independent_review`; the package README calls this historical. A one-line pointer to the accepting review would avoid the appearance of an open item.
- Theorem thm:v6-962-polynomial says "under that theorem's arithmetic hypotheses"; as with Theorem thm:quartic, naming them in the statement (RC, Theorem hyp:o8-height, Theorem thm:v5-simultaneous-primes) would make the conditional reading unambiguous.

Discrepancies: none in any recomputed quantity or re-derived step.

Next checks.

1. An audit of the appendix "Simultaneous primes with coprime adjacent norms" by someone who works with the Green–Tao machinery, or alternatively a decision to cite Sun's Theorem 4.1 and Frei–Tinková Theorem 1.4 and carry that dependence explicitly.

Either way the 962 result currently inherits an input whose published status is "to appear".

2. The restricted converse remains the main obligation for all three polynomials.
3. Read the remaining local lemmas of the 962 section (two anchors, height transfer, unramified log, reduced group cardinalities) with the line-by-line method; the five I re-derived were correct.
4. Recompute the Version 5 table independently if that polynomial is to be cited on its own; Version 6's chain only pins it by hash.

Update: Version 15 (commit 511529b, 10 Nov)

What changed, and what this pass covers

Audited head: `511529bbe9d635194f0c9d925709e77a5d3afa64` (10 Nov 2026), 59 commits after the head of the previous update (`f8722656…` , Version 6). The manuscript is now `research/version15/snapshots/v15-222-004/manuscript/quartic_v15.tex` : 39,896 lines and 546 pages, against 19,034 lines for `quartic_962.tex` . Its SHA-256 (`44ba3a8e…`) and the PDF's (`c6ec627e…`) match the repository's build record. `lean/` has not changed since the last audit; the main declaration still contains `sorry` .

The versions in between, from the repository's own history:

Version	Content
7	Four further audits of the Gaussian appendix, with repairs
8	Revised abstract
9	Odd-product quartics: 523 variables with a symbolic constant, 540 with decimal coefficients
10	Response to nine points raised by an outside reviewer in GitHub Issue 2
11	Repair of the application of Kato's Euler system, after the same reviewer's follow-up
12, 13	Quartics in 498/514 and 394/411 variables

Version	Content
14	Quartics in 299 variables (symbolic constant) and 314 variables (every coefficient in decimal)
15	A family of quartics in 222 variables, depending on an integer parameter

The manuscript now states three headline theorems: the original polynomial in 292,890,231,905 variables, the 314-variable decimal quartic, and the 222-variable family. Its text divides as follows.

Lines	Content	What I did in this pass
1–20,351	The Version 6 text with 141 changed passages (+2,462 / −1,145 lines)	Read every changed passage outside the Gaussian appendix; in that appendix read the sieve changes in full and sampled the rest. Unchanged text was not re-audited.
20,352–23,825	Versions 9–14: the chain 962 → 523 → 498 → 394 → 299, then the lift to 314	Re-derived the first Version 9 lemmas; recomputed the 314 table and checked the lift. The sections for Versions 12–14 were not audited.
23,826–30,051	New part: an internal proof of the inverse theorem for Gowers norms	Not audited.
30,052–33,139	Version 15 arithmetic: contact height, amplification, aggregate height, formal logarithms, prime selection, the 248-variable baseline	Re-derived, except the completeness half of the baseline, which I followed without reconstructing.
33,140–37,078	Reductions 248 → 238 → 233, with explicit 238- and 233-variable systems	Read; re-derived the one new arithmetic lemma; checked the explicit 233 system against the package by machine.
37,079–39,698	Reductions 233 → 222 and the explicit 222-variable quartic	Checked every stage by machine with my own code.

Two limits on this pass that I could not remove. First, the sandbox refused to execute code from the repository, so I did not run the package verifier (`quartic-222/verify.py`) or the repository's tests. Every computation reported below is a script of mine that reads the repository's JSON and TeX as data. Second, GitHub's issue, pull-request and check pages were not reachable from here, so I know the outside review threads and the PR 26 protocol evaluation only through the repository's own records of them.

The Version 15 claim

Statement. Theorem `thm:v15-main` : for $N \in \mathbb{N}$ put A_N = 149^(2N+1) and P_N = Σ w_i·Q_{i,N}², where Q_{1,N}, …, Q_{196,N} are explicit polynomials of degree at most 2 in 222 coordinates and the w_i are positive integer weights. There is an integer N_0 such that, for every $N \geqslant N_0$ and every rational x, x is an integer exactly when $P_N(x, z_2, \ldots, z_{222}) = 0$ has a rational solution. The theorem adds that every P_N has degree 4 and content 1, that for $N \geqslant 53$ it has 2,277 monomials and largest coefficient $A_N^2 + 1$, and that N_0 is ineffective.

How the proof is organized. The construction lives on the curve $E: y^2 = x^3 - Dx$ over $K = \mathbb{Q}(\sqrt{2})$, with $D = (3 - \sqrt{2})^2$, and starts from a 248-variable system that the manuscript credits to José Brox. The input x enters through $\beta = 2504 + 51x^2$ and a scalar $R = A_N \cdot r_4/(1 + 149^2 \cdot r_4)$, where $r_4 = \beta(\beta - 1)(\beta - 2401)(32\beta + 343)$. Soundness (a rational x with a witness is an integer) combines three things: a forced divisibility of a scalar index by a high power of 149 and by the cube of a radical $M_5(\beta)$ built from the primes at five "contact" values; a new height bound for that index, $h(a) \leqslant C_3 \cdot M_5(a)^3 \cdot (1 + \log M_5(a))$ with an explicit $C_3 <$ 2^330770; and a bound for the total canonical height of several multiples of the base point in terms of the height of a single trace. Completeness (every integer has a witness) uses simultaneous prime values of Gaussian norm forms and the realization of points with prescribed poles, both inherited from the earlier versions. Twelve reduction steps then bring 248 variables down to 222.

Where the ineffectivity comes from. One place only. The aggregate height bound (Section `v15:aggregate`) has the shape $\Sigma\, \hat{h}([n_i]P) \leqslant M \cdot h(\text{trace}) + M_0$, and M_0 comes from Von Buhren's theorem (J. Théor. Nombres Bordeaux 29 (2017), Théorème 1.1), whose additive constant is not effective. I fetched the paper. Its statement is the one the manuscript uses: an explicit multiplicative constant, an ineffective additive constant, and an exceptional set made of translates of positive-dimensional abelian subvarieties. The manuscript's proofs that no such translate occurs here check on re-derivation. Everything else in the soundness argument is explicit.

N ⩾ 53 concerns coefficient size only. It is the point from which $A_{N^2} + 1$ exceeds the largest parameter-free coefficient, which has 1,521 bits. It says nothing about N_0, and the theorem, the README and the release notes all say so; I found no place where the two thresholds are conflated. For a sense of scale: by my reading, the explicit half of the soundness argument on its own needs 149 raised to about 9N/2 to exceed $7 \cdot C_3$, which already puts N near ten thousand before the ineffective constant is added. So no member of the family that can be written down today is known to define $\mathbb{Z}$. That is the manuscript's own position.

The abstract's sentence about C. The abstract says the second quartic's "coefficients are explicit polynomials in a positive integer C, for which we have no effective upper bound." The repository records this as a requested edit of the earlier sentence ("…we have no effective bound for a member known to define the integers"). The new sentence is compatible with the theorem, but read alone it suggests that C is one particular number of unknown size. What the theorem gives is that every $C = 149^{(2N+1)}$ with $N \geqslant N_0$ works; what is unknown is how large N_0 is. If the abstract should carry that, wording such as "…in a positive integer parameter C, which must be taken sufficiently large; no effective bound for how large is known" would do it.

The 222-variable polynomial: independent recomputation

The package is `research/version15/quartics/quartic-222/`. It holds a raw and a sparse presentation; the sparse one is the polynomial of the theorem. I expanded $\Sigma\, w_i \cdot Q_i^2$ from the residual lists with my own code, treating A as a formal parameter, and compared the result with the published coefficient tables.

Claim	Result
196 residuals of degree at most 2 with integer coefficients	Confirmed; every row is primitive
Positive integer weights	Confirmed. Weight 2, except rows 40, 53, 119, 121, 153, 155, 169, 171 (weight 4); 57, 71, 116, 132, 150, 166 (weight 1); 117, 151, 167 (weight 8); 133, 183 (weight 6). Sum 428
$\Sigma\, w_i \cdot Q_i^2$ equals `sparse222-polynomial.json`	Equal term by term (2,278 terms with the parameter separated)

Claim	Result
2,277 coordinate monomials, degree 4, all 222 variables occur	Confirmed; 1,545 of the monomials have degree 4
Content 1	Confirmed for every N: the coefficients that do not involve A already have gcd 1
A enters through one residual	Confirmed: row 71, which reads $z_{120} + 149^2 \cdot z_{108} \cdot z_{120} - A \cdot z_{108}$. After squaring, A occurs in three monomials: z_{108}^2 with coefficient $A^2 + 1$, $z_{108} \cdot z_{120}$ with $-2A$, and $z_{108}^2 \cdot z_{120}$ with $-44402 \cdot A$
Largest coefficient is $A_N^2 + 1$ for $N \geqslant 53$	Confirmed, and it fails at N = 52. The largest parameter-free coefficient has 1,521 bits and is below 149^214
The input occurs in one residual	Confirmed: row 9, which reads $2501 + 51 \cdot z_1^2 - 51 \cdot z_{55}$
Raw presentation: 2,961 coordinate monomials	Confirmed, with the same sum-of-squares identity

The monomial count is 2,277 for every N I tried (0, 1, 2, 10, 52, 53), so the theorem's restriction to $N \geqslant 53$ is needed only for the height.

The printed equations are the table. I parsed the 196 displayed equations (Q1)–(Q196), the eight constants $\kappa_1, \ldots, \kappa_8$ and the weight table from the manuscript's TeX source and compared them with `sparse222-ir.json`. No row and no weight differs. The polynomial in the theorem, the polynomial printed in the manuscript and the polynomial in the package are therefore one object.

The 314-variable decimal quartic. For Theorem `thm:v15-decimal` I recomputed the same kind of facts from `quartics/quartic-314-fully-expanded.txt`: 6,672 monomials in 314 variables; degree 4, with 4,328 monomials of degree 4, 1,705 of degree 3, 580 of degree 2, 58 of degree 1 and a constant; content 1; file size 264,430 bytes; largest coefficient of exactly 6,496 bits, hence below 2^6496 as stated. The table equals the sum of squares of the 294 residuals in `compact314-ir.json` once each is scaled to a primitive integer row. I also checked the lift from 299 to 314 variables against that file. Fourteen coordinates carry a chain of constants: 3^2049, then eleven steps $c \mapsto 3^{2048} \cdot c^2$, then two squarings, ending at C = 3^(2^25). Exactly two residuals use the last of them, and one further coordinate holds the quadratic form Q of the manuscript. Each constant is determined

uniquely, so the lift introduces no sign ambiguity. This confirms the finite description of the 314-variable polynomial. It does not touch the arithmetic of Versions 12–14 on which that theorem rests, which I did not audit.

Scripts: `/home/claude/audit/v15/check222.py`, `check222_tex.py` and `check314.py` in the same directory.

The reduction chain to 222 variables

The manuscript reaches 222 variables in twelve steps: 248 → 245 → 240 → 239 → 238 → 237 → 235 → 233 → 232 → 226 → 224 → 223 → 222. It prints three systems in full, with 238, 233 and 222 variables. The package starts at 233 and stores a table for each later stage.

From the printed 233-variable system to the printed 222-variable system: checked by machine. For each stage I wrote my own substitution code and compared the systems before and after as sets of quadrics, in exact rational arithmetic. I did not use the repository's scripts. Where the manuscript prints a certificate, I checked it by expansion.

Stage	What changes	What I verified
Printed 233 system and the package table	The manuscript prints the 233 system in sparse coordinates; the package's chain starts from the raw system	Applying the nine coordinate changes printed in the manuscript to the 207 raw rows gives exactly the span of the 207 printed equations (both of rank 207). The input coordinate is untouched
233 → 232	The factor $2\beta - 1$ is dropped from the contact product; one coordinate and one row go	Only two rows used the deleted coordinate; the other 205 are unchanged; the new row is $x_{123} = r_4$ identically. This step changes the arithmetic construction and is not a bijection of witnesses (see below)

Stage	What changes	What I verified
232 → 226	Four ratio coordinates $z_0, \ldots, z_3$ over K are replaced by one common ratio T	The 16 old rows are the components of N_r·z_i + D_r·L′_i and E_i + K_i·z_i; the 10 new rows are the components of D_r·T − N_r and E_i·T − K_i·L′_i; 190 rows are unchanged; the printed identity holds
226 → 224	The coordinate g is eliminated by a polynomial formula; then the quadric Q_5 is dropped	With my own arithmetic over $\mathbb{Q}(\sqrt{2})$: the printed quadrics $Q_0, \ldots, Q_5$ are the package rows; the printed certificates for the two reference equations hold after the substitution and use only Q_0, Q_1, Q_3; the curve identity holds; all six quadrics vanish on the old solution set (160 generic points, exact); Q_2 is linear in w with the printed coefficient M; 192 rows are unchanged and none uses g
224 → 223	Rescaling by q and R; six square outputs deleted; five helper coordinates added	Of the 198 old rows, 184 become single new rows up to a constant and a monomial in q and R; six become identically zero; six detector rows become the stated combinations $Q_1 \cdot \mathrm{det}_0 - Q_0 \cdot \mathrm{det}_1$; two definitional rows are literal. The ten rows that give q ≠ 0 and R ≠ 0 are untouched
223 → 222	Two trace helpers G and Z are replaced by one quotient	With $G = R^2 \cdot T$ and $Z = Q_1 \cdot T$, one old row vanishes identically and the other 196 are the new rows

Stage	What changes	What I verified
Raw 222 → sparse 222	Eight affine coordinate changes, row operations, positive weights	The eight changes, parsed from the manuscript, are invertible and fix the input; the 196 transformed raw rows and the 196 sparse rows span the same space (rank 196)

The domain conditions. Several stages divide by a quantity, so the correspondence of solution sets needs that quantity to be nonzero on every rational solution. I traced each one.

- $q \neq 0$ and $R \neq 0$ (stages 224 → 223 and 223 → 222). I re-derived both from the actual rows of the raw 222 system. Four rows give $R = A \cdot r_4/(1 + 149^2 \cdot r_4)$ with $r_4 \geq 1$, so $R > 0$. Five more give $(1 + q) \cdot P_5(r) = r^5$ with r a rational square, where P_5 is monic with 5-integral coefficients and $P_5(r) \equiv (r - 1)^2(r - 4)^3$ modulo 5. Then $q = 0$ is impossible, by reduction modulo 5 when r is 5-integral and by the leading term otherwise.
- T is a 149-adic unit (232 → 226); $a + cf$ and $\beta_0 S + \alpha$ are nonzero, and the coefficient M is nonzero (226 → 224). These follow from the 149-integrality and unit norms of the printed constants, which I checked, together with the valuations $v(f) = m > 0$, $v(S) = -m/2$, $v(g) = m/2$ at the inert prime 149. Those valuations come from the scalar part of the soundness argument. I re-derived that argument in the 248-variable baseline and in the 237-variable criterion; I did not re-trace it row by row in the reduced systems.
- $f \neq 0$ (226 → 224) is direct: $f = 0$ forces a nonzero constant to vanish, as printed.

The step 233 → 232. Here the scalar attached to the input changes, so both directions of the theorem are argued again and not transported. For soundness the facts used are these: the four residues $(\theta - 2504)/51$ at the finite contacts θ are nonsquares modulo 149, so r_4 is a 149-adic unit for 149-integral x and $v_{149}(R)$ is $2N + 1$ or $2N - 1$; and the pairwise differences of the contacts are supported on {2, 3, 5, 7}, so a contact prime above the cutoff divides exactly one factor. I verified these numerically and re-derived the conclusion that $149^{(\lfloor 9(2N-1)/4 \rfloor + 1)} \cdot M_5(\beta)^3$ divides the scalar index. For completeness I verified the local facts the text uses: $v_2(r_4) \in \{2, 3\}$, $r_4 \equiv 1 \bmod 3$ and $r_4 \equiv 13 \bmod 17$, hence $1 + 9792/R$ is a square in $\mathbb{Q}_2$, $\mathbb{Q}_3$ and $\mathbb{Q}_{17}$, and the reduced denominator of $R(R + 9792)$ is a square. The construction of the witnesses themselves is by reference to the predecessor's argument, which I followed without reconstructing.

Upstream of 233. I read the steps 248 → 238 → 233 as text. I re-derived the three-residue scalar criterion (238 → 237), including its exponent ⌊9(2N − 1)/4⌋ + 1. I did not compare the printed 238 system with a table or re-run those maps.

Scripts, all in `/home/claude/audit/v15/`: `join233.py`, `stage_checks.py`, `stage_B.py`, `stage_C.py`, `stage_D.py`, `sparse_span.py`, `numeric_claims.py`.

New arithmetic and the status of the earlier dependencies

New in Version 15. "Verified" below means that I reconstructed the argument and its constants from the text. A theorem cited inside an argument is noted separately.

Item (section)	Status	Notes
Faltings height of a Jacobian bounded by evaluating integral forms (`v15:height`)	Verified	Trace-dual lattice, integrality of the evaluations, a mean-value estimate on a disc, Hadamard's inequality. The discriminant lemma uses a bound on the different that the manuscript cites as Javanpeykar, Prop. 4.1.3; I did not check that statement against the source
Modular case: h_F(J_H) ⩽ (d_H/2)(log 12 + log N + 2 log d_H) + g_H·log(2/√π)	Verified	Including the explicit model with j = 1728t and the degree formula

Item (section)	Status	Notes
Contact-height theorem: $h(a) \leqslant C_3 \cdot M_5(a)^3(1 + \log M_5(a))$, $C_3 <$ 2^330770	Arithmetic verified in full	It rests on a "modular realization" lemma extracted from the proof of the earlier height theorem. I traced the extraction (the bounds $[L':\mathbb{Q}] \leqslant 2304 \cdot \varphi(R)$ and level $\leqslant$ 210^300·R^2, the isogeny-factor argument, the nebentypus) and found it consistent. The geometric expansions behind it (level-3 marking, rigidity, canonical models) I followed without reconstructing. Cited and not checked against the sources: Gaudron–Rémond 2023 (Lemma 16.2(1), Thm 1.9(1)), Rémond 2022, Carayol 1987, and modularity at p = 5 as quoted from Calegari's ICM 2022 survey
Amplification over disjoint contact blocks (`v15:amplification`)	Verified	Elementary
Aggregate trace height (`v15:aggregate`)	Verified	The no-translate theorems, the independence of P and its conjugate over $\mathbb{Z}[i]$ modulo torsion, and the infinite order of P all check. Von Buhren's theorem was checked against the paper
Truncated formal logarithms (`v15:logs`)	Verified	Differential equation, coefficient recurrence and valuation bound, tail estimate, positivity
Selection and avoidance lemmas (`v15:selection` , `v15:dependencies`)	Elementary parts verified	The analytic input is the Gaussian simultaneous-primes theorem of the appendix

Item (section)	Status	Notes
248-variable baseline, soundness (`v15:base248`)	Verified step by step	Scalar extraction and its three valuation bounds, the forced 149-adic depth, $M_5(\beta)^3$ dividing the index, the residue lemma, integrality transfer, the two height contradictions, the Lucas lower bound, and the last step from $n = \beta$ to $x \in \mathbb{Z}$
248-variable baseline, completeness	Followed, not re-derived	Prime patterns and pole realization; depends on the inherited theorems below
Three-residue scalar criterion, 238 → 237 (`v15x:parity`)	Verified	
Six short supporting sections (unary theta series, theta transport, Tate-module and marking descent, Picard norm, conductor-one CM reduction)	Read once, not re-derived	The last replaces the earlier citation of Cornut–Jetchev by an internal argument

I found no error in any item marked verified.

Inherited dependencies. The 222-variable theorem still rests on the theorems reported on earlier in this document: the restricted converse (with Kato's Euler system at $p = 2$), pole realization, the identity $2E(K) = \mathbb{Z}P$, and simultaneous prime values of Gaussian norm forms. Each has been edited since Version 6. What I can say about the edits:

- **Kato's Euler system (Version 11).** The outside reviewer's follow-up identified a real gap: the family of classes was built on one index set, Kato's 1999 theorem was applied on another, and the limiting class was not identified. The repair (lines 13,780–13,894) defines the classes on the larger index set, verifies both cases of Kato's norm relation there, and identifies the class. I read it and it is coherent. I did not check it against Kato's papers. My audit of 4 November had passed this step as coherent given the cited inputs and did not notice the mismatch; the project's internal reviews had accepted it as well.
- **Restricted converse, other edits.** Parity now cites Dokchitser–Dokchitser, Theorem 1.4, over $\mathbb{Q}$. There is a new determinant-specialization lemma and several clarifications. Read; no objection.

- **Height section of the original construction.** The de Rham descent appendix was removed. In its place a lemma (`lem:qm-modularity-descent`) applies the p = 5 modularity theorem as quoted in Calegari's survey, after checking weight 2 and a finite twist. I checked the lemma's logic; the cited theorem is at citation level.
- **Gaussian simultaneous primes.** The sieve estimates were expanded (Rankin's bound, successive minima, uniformity in the cutoff). I read those in full and found them correct. The inverse step no longer uses Dodos–Kanellopoulos; it uses Green–Tao's relative inverse theorem together with the manuscript's own proof of the Gowers inverse theorem. I only sampled the remaining changes.
- **Gowers inverse theorem (lines 23,826–30,051).** About 6,200 new lines. The text re-assembles the Green–Tao–Ziegler inverse theorem in the form it needs and replaces several steps of the published proof that the project regards as defective. I did not audit it. The theorem itself is published and later proofs by other routes exist, so whether this part is a dependency or a citation is a choice; see the findings.
- **Degree lower bound.** The unirationality input is now Colliot-Thélène–Salberger 1989, Prop. 1.3, in place of Kollár 2002. Not checked against the source.

Versions 9–14. These sections support the 314-variable theorem, not the 222-variable one. In Version 9 I re-derived the chart-replacement lemma and proposition and the odd-product height identity. I did not audit the sections for Versions 12, 13 and 14.

Review record, protocol and repository integrity

Outside review since the last audit. The repository records two GitHub issues from outside readers. Issue 2 raised nine points; Version 10 answers them, and the reviewer's follow-up of 7 November led to the Version 11 repair of the Kato step described above. Issue 3 asked for exact reproducibility details of the finite checks of the original construction. The reply documents them, reports a byte-for-byte regeneration of the original polynomial, and invites the contributor's conditional Lean proof (eight explicit hypotheses) as a pull request. I could not open the issues themselves from here.

The acceptance policy. On 9 November the repository adopted a policy (`docs/research-review/ASTRA-REVIEW-POLICY.md`) under which only designated agents with the model identifier `gpt-6-astra` may supply the judgments that count as acceptance, the accepting reviewer must not be an author of what is accepted, and enforcement runs in a controller outside the pull request. The policy itself says that shared model identity does

not establish independent reasoning. I read it as a rule about who may sign. It does not bear on whether a given argument is correct, and I did not use any acceptance record as evidence in the sections above.

What the Version 15 records claim.

- `research/version15/protocol/current222/` holds 193 theorem applications, 505 declared dependency edges, 1,187 source uses, 1,167 connections and 32 formal findings, of which 30 are marked repaired and two, about packaging, out of scope. The directory describes its files as unsigned preparation records that do not supply pull-request identities or authenticated acknowledgments, and it points to the protocol evaluation of PR 26 on GitHub as the actual acceptance. That evaluation is not in the tree and I could not read it.
- The manuscript review (`evidence/222-manuscript-review/REVIEW.json`) records the decision `accepted_at_new222_mathematical_integration_and_printed_equation_scope` and states its own limit: the acceptance is "conditional on the retained accepted arithmetic foundation" and "is not a fresh independent proof audit of the entire 546-page work."
- Four of the recorded findings show the process catching real defects before release: an index error in an earlier table of the 233/235 stage (corrected; the erroneous version was not selected); a reversed clause in the contract for the modular-realization lemma (repaired); an acceptance withdrawn because the reviewer had written part of what was reviewed; and three sentences that presented the 233-variable predecessor as the final theorem. I confirmed that the current text no longer does that.
- Each stage of the 222 package has its own review directory. I did not evaluate those reviews. The checks in the previous sections are independent of them.

Statuses a reader will trip over. Several files linked from the Version 15 index still carry a pre-acceptance status because they are preserved byte for byte: `ACTIVE-REVIEW-RESOLUTION.json` says "independent semantic review pending"; the stage notes under `quartic-222/provenance/` say "author proposal" or "pending independent review"; the manuscript's `BUILD.json` says "visual delta QA pending". The READMEs explain that these are historical. Together with the point about PR 26, this means the tree alone does not let a reader confirm that the 222 construction passed the protocol.

Integrity.

- The manuscript and PDF hashes match `BUILD.json`. The current snapshot differs from the previously selected one in one line of the abstract (`CHANGES.json`), the sentence discussed above.
- In the 222 package, all 11 stage manifests match the hashes in `CHAIN.json`, and all 215 files listed in `SOURCE-COPIES.json` match their recorded hashes.
- The root `SHA256SUMS` verifies for all 32 entries present in the tree; the nine release archives it lists are not in the tree, as before. Its entries stop at Version 10, so the files of Versions 11–15 are covered only by the per-version manifests.
- The 314-variable table is byte-identical in the three places where it appears.
- `lean/` is unchanged.

Findings for this update

Short answer. The finite claims hold exactly. I recomputed the 222-variable quartic and the 314-variable quartic from their residual lists; the equations printed in the manuscript are the package's table; and every algebraic step from the printed 233-variable system to the printed 222-variable system checks by machine, with the domain conditions accounted for. In the new Version 15 arithmetic I found no error on re-derivation, and the theorem's ineffectivity has a single, correctly identified source. The 222-variable theorem remains conditional. I mean that in the sense of resting on parts I have not validated, not in the sense of a failed inference. Those parts are the inherited theorems as edited since Version 6, a 6,200-line internal proof of the Gowers inverse theorem, and cited results that I did not check against their sources.

One fact from the history deserves more weight than anything I computed. Version 11 exists because an outside reader found a real gap in the application of Kato's theorem, in a step that the project's internal reviews had accepted and that my own audit of 4 November had passed. That is direct evidence about what model review, mine included, does and does not catch in the inherited arithmetic, and it is why I order the next checks as I do. As before, what follows is model review; it is neither a referee's report nor a formal proof.

Part of the claim	Status	Basis
The 222-variable polynomial is as described: variables, residuals, weights, monomials, degree, content, height from N = 53	Verified	My own expansion; printed equations compared with the table
The 314-variable polynomial is as described, including the lift from 299 variables	Verified	My own expansion; constant chain inspected
Printed 233 system → printed 222 system preserves the defined set	Verified, given the 149-adic valuations supplied by the scalar argument	My own code for every stage; domain conditions traced
Contact height, amplification, aggregate height, formal logarithms, three-residue criterion	Verified	Re-derived
248-variable baseline: soundness	Verified	Re-derived step by step
248-variable baseline: completeness; steps 248 → 233; witness construction after 233 → 232	Followed, not re-derived	Read; numerical claims checked
Geometric input to the modular realization lemma; six short supporting sections	Followed, not re-derived	Read once
Von Buhren's theorem as used	Checked against the source	Paper fetched

Part of the claim	Status	Basis
Other cited results: Javanpeykar Prop. 4.1.3; Gaudron–Rémond; Rémond; Carayol 1987; modularity at p = 5 via Calegari's survey; Kato's 1999 paper and his Astérisque paper; Colliot-Thélène–Salberger; Ribet 1990; Dokchitser–Dokchitser	Not checked against the sources	Citation level
Inherited theorems as edited: restricted converse with the Kato repair, pole realization, 2E(K) = ℤP	Changed passages read; nothing else re-audited	My earlier coverage is described in the previous sections of this document
Gaussian simultaneous-primes appendix as edited	Sieve changes read in full; other changes sampled	
Internal proof of the Gowers inverse theorem	Not audited	
Arithmetic of Versions 12–14, which supports the 314-variable theorem	Not audited	
Repository verifier and tests	Not run	The sandbox refused repository code
Protocol acceptance of Version 15 (PR 26)	Not verifiable from here	Recorded on GitHub only

Needs clarification.

- The abstract's sentence about C. It is compatible with the theorem but reads as if C were one number of unknown size.
- The acceptance of the 222 construction is recorded on GitHub, while the files in the tree describe themselves as preparation records or still say "pending". An in-tree copy of the PR 26 evaluation result, or one sentence in `research/version15/README.md` naming the authoritative record, would let a reader of the archive see the status without the web.

- The root `SHA256SUMS` stops at Version 10. Either extend it, or say in the README that later versions are covered by their own manifests.
- The manuscript proves the Gowers inverse theorem internally because the project regards parts of the published proof as defective. If that judgment is part of the claim, the specific defects and their repairs need a reader who knows that literature. If it is not, the part could be reduced to a citation of the published theorem in the exact form used.

Discrepancies. None between the manuscript, the tables and my recomputation.

Next checks, in the order I would do them:

1. A specialist reading of the Kato step as repaired in Version 11, against Kato's two papers. This is the one place where a gap is known to have passed model review.
2. Decide the status of the inverse-theorem part (cite or prove). If it stays as a proof, have it audited separately.
3. An independent reconstruction of the completeness half: the 248-variable baseline, the witness construction after the four-contact change, and the pole-realization inputs it uses.
4. A machine comparison of the printed 238-variable system with a table, and of the maps 248 → 238 → 233, to extend the machine-checked chain back to the baseline.
5. The arithmetic of Versions 12–14, if the 314-variable theorem is to be relied on separately.
6. Source checks of the cited results listed in the table.
7. Run `verify.py` and the repository tests in an environment that allows it.

Second pass on Version 15: the six open items (10 Nov)

Scope: the six items and how each was approached

After the first pass Astra listed six further things to audit. This second pass takes them one at a time. The sections below follow Astra's numbering; I worked on them in the order 4, 1, 6, 3, 5, 2.

Three outcome labels are used throughout. **Verified** means I re-derived the inference or reproduced the computation with my own code. **Followed** means I read the argument and found no gap, without re-deriving it. **Not audited** means I did not examine it. A result is

called conditional on something when it depends on that thing and that thing is not verified here. That describes coverage; it does not mean an inference failed.

Item	What I did	Outcome
1. Repaired Kato application	Compared it with Kato's Astérisque 295 paper and re-derived the repair steps. Could not obtain Kato's 1999 Kodai paper	Astérisque side confirmed where the text was legible. The 1999 paper is not checked
2. Internal inverse theorem	Read and re-derived eight of seventeen sections and part of a ninth; checked four claims about the published sources	Those parts verified. Eight sections not audited
3. Completeness and pole realization	Re-derived pole soundness, simultaneous realization and the baseline completeness argument; checked the later changes, and the charts at exact sample points	Verified. Rests on three results not re-audited
4. Chain 248 → 238 → 233 by machine	My own code transports every row of the 248-variable table through the manuscript's dictionaries to the printed 238-variable table, then to 237, 235 and 233	Verified at table level. Joins the first-pass chain from 233 to 222
5. Arithmetic behind the 314-variable quartic	Read and re-derived Versions 9, 12, 13 and 14; machine-checked the finite identities; compared the stored system with the text	Verified. Rests on results proved earlier in the manuscript
6. Cited theorem applications	Compared the quoted statements with the sources I could reach	Six sources match. Two could not be read. Seven not attempted

Limits that applied to the whole pass:

- I did not execute the repository's code. Every check uses my own scripts, which read the repository's tables and the manuscript as data.
- Kato's 1999 paper is hosted on two sites that refuse automated access, and I did not go around that. Ribet's 1990 paper was available only as a scan with no text layer.
- Your computer was not linked to this session, so I could not read local copies of either.

Item 1: the repaired Kato application against the two papers

Result. Half done. Against Kato′s Astérisque paper (2004) the statements the manuscript uses are there, as used. Against Kato′s Kodai paper (1999) I could not check anything, because I could not obtain it. Given the reading of the 1999 paper that the outside reviewer and the project each recorded, the repair is correct.

Access. The Astérisque paper is on Numdam. Its text layer is OCR and many formulas are garbled, so "confirmed" below means confirmed where legible. The Kodai paper is on Project Euclid, which returned a bot-protection page, and on J-STAGE, whose robots policy refuses automated retrieval. Your computer was not linked to this session. I did not work around either site.

Astérisque 295, statement by statement.

Used in the manuscript	Place	What the paper says
The smoothed classes at level m	(8.1.3), with (8.1.2)	Elements in $H^1(\mathbb{Z}[1/p, \zeta_m], V_{O_\lambda}(f)(k-r))$, under prime(cd) ∩ S = ∅, (cd, 6) = 1, (d, N) = 1. Agrees
The integral groups are cohomology of j_*	§8.2	H^q(R, ·) abbreviates H^q(R, j_*(·)), defined as an inverse limit over finite coefficients. Agrees
Norm relation with a product over newly added primes only	Prop. 8.12	For m dividing m′, S ∪ prime(m′) ⊂ S′ and prime(cd) ∩ S′ = ∅, the norm of the m′-class with S′ is a product over ℓ in S′ \ S of Euler factors times the m-class with S. Agrees; the factor′s formula is illegible in the scan

Used in the manuscript	Place	What the paper says
Smoothing scalar $(c^2 - c)(d^2 - d)$	(4.2.4), Thm 6.6	$(u, v) = (r + 2 - k, r)$ when $r' = k - 1$, hence $(1, 1)$ here; the constant is $(c^2 - c^{\wedge}u \cdot \chi(c))(d^2 - d^{\wedge}v \cdot \chi(d))$ in the $SL_2(\mathbb{Z})$ case with $c \equiv d \equiv 1 \bmod N$. Agrees
Definition of an Euler system	§13.1	A finite set Σ containing p and the ramified primes; indices m with $\mathrm{prime}(m) \cap \Sigma = \{p\}$; z_m in $H^1(\mathbb{Z}[\zeta_m, 1/p], T)$; norm relation with $P_\ell(\ell^{-1}\sigma_\ell^{-1})$, where $P_\ell(t) = \det(1 - \mathrm{Fr}_\ell \cdot t \text{ on } T)$ with arithmetic Frobenius. Agrees
Theorem 13.4	p. 226	Hypotheses (i)–(v) as the manuscript lists them. (1) H^2 is torsion. (2) The length bound at height-one primes not containing p. (3) The same at all height-one primes, under residual hypotheses and $p \neq 2$. The restriction $p \neq 2$ appears only in (3). Agrees
Rank and the group $H^2(T)_0$	(12.2.1)–(12.2.2), before 13.4	Finite generation, the rank formula, and $H^2(T)_0$ as the kernel of the map to local cohomology at p. Agrees
Nonvanishing inputs	Thm 13.5(2), 13.6, Prop. 13.7	Rohrlich's finiteness of vanishing twists, Ash–Stevens generation, nonvanishing at every height-zero prime. Agrees

Two observations.

First, the Astérisque paper does not prove Theorem 13.4. It says "By [Pe4, Ru4, Ka4], we have Theorem 13.4", the three references being Perrin-Riou, Rubin, and Kato's 1999 paper. As printed, parts (1) and (2) carry no restriction on p, and the index set is the one that the manuscript's family (K8) already satisfied before the repair. So the published statement

the manuscript needs is, on its face, in the 2004 paper for p = 2 as well. The 1999 paper matters because it is the one cited source that treats p = 2, and its definition of an Euler system uses a different index set. That is exactly the gap the reviewer found.

Second, the repair itself. Take the 1999 definition as the reviewer and the project record it: classes at every m divisible by N_K and prime to N′_K; a norm relation that is the identity when ℓ divides m and $P_\ell(\ell^{-1}\sigma_\ell^{-1})$ when it does not; and a limit class defined by norms from the levels $N_K \cdot 2^n$. I re-derived the three steps of the repair against Prop. 8.12:

- The classes exist for every m prime to c·d_s, because the only condition is that prime(c·d_s) avoids the support prime(2·D·m·N).
- If ℓ divides m, the supports at Dmℓ and Dm coincide and the product in Prop. 8.12 is empty. If ℓ does not divide m then, since 2Nd divides m, ℓ is a new prime and exactly one factor appears; pushing it through the twist by χ and the corestriction turns σ_ℓ^{-1} into $\chi(\ell)\sigma_\ell^{-1}$, which gives the Euler factor of the twisted representation.
- At the levels $D \cdot N_K \cdot 2^n$ and $D \cdot 2^n$ the supports are both prime(2Nd), so the norm between them carries no factor, and the limit class is the retained class y at the levels 2^n.

I found no error in these steps.

What is still open. Someone with the Kodai paper in hand should confirm three things: the index set and relation (0.1.1) of its §0.1; the definition of the class in §0.7; and conditions 0.6 with the statement of Theorem 0.8 for p = 2 at height-one primes not containing 2, including that the unipotent element is required in the Galois group of the maximal abelian extension. If you attach the PDF I will do it.

Item 2: the internal proof of the Gowers inverse theorem

Result in brief. I verified eight of the seventeen sections and part of a ninth, about 2,700 of 6,240 lines, and found no error in them. I also checked four of the manuscript's claims about defects in the published sources against those sources, and all four are right. I did not audit the other eight sections, which hold the hard core of the argument, so this audit does not establish the internal proof as a whole. The statement being proved is a published theorem, and the last part of this section sets out what citing it would require.

Superseded in part by the third pass. The eight sections listed below as not read have since been read and verified; see "The remaining inverse-theorem sections" in the third pass. The coverage table, the first open point and the closing paragraph of this section describe the state after the second pass only.

What this part of the manuscript is

The part runs from line 23826 to line 30064. Its target, called GI(k), is: for fixed k ⩾ 3 and δ > 0 there is a finite family of nilmanifolds of step at most k such that every 1-bounded f on [N] whose U^(k+1)[N] norm is at least δ correlates with a Lipschitz function along a linear orbit on one of them. This is Theorem 1.3 of Green, Tao and Ziegler (Annals of Mathematics 176, 2012). The manuscript uses it for k = 20 and k = 39.

The part is not a self-contained proof. It is the published Green–Tao–Ziegler argument with individual steps replaced or re-derived, plus a record of how the pieces join. Everything that is not replaced is taken from the published paper by section and page, so the published paper has to be open to follow it.

Coverage

Section	Lines	What I did	Outcome
Assembly of the inverse-theorem input	23828–24414	Read in full, re-derived	Verified
GI(2) base from the published U^3 theorem	24415–24700	Read in full, re-derived	Verified, given the published forward theorem
Tautification and shifted nilcharacters	24701–24937	Read in full, re-derived	Verified
Replacement for Lemma 13.2	24938–25317	Read in full, re-derived	Verified
Difference decomposition, additive quadruples	25318–25601	Read in full, re-derived	Verified, given the published selection and Fourier lemmas
One-variable factorization and linearization	25602–26076	Not read	Not audited
Equal-scale factorization, bias and torsion	26077–26576	Not read	Not audited
Sections 9–11 at every rank step	26577–27137	Closing subsections only	Not audited

Section	Lines	What I did	Outcome
Petal quotient and degree–rank construction	27138–27672	First two subsections re-derived (to 27297)	Filtration lemma and integral coordinate lemma verified; rest not audited
Section 12 semidirect-product join	27673–28169	Not read	Not audited
Degree conversion and rank iteration	28170–28435	Closing subsections only	Not audited
Scaling symbols and multilinearisation	28436–28874	Not read	Not audited
Plain additive-quadruple implication	28875–29024	Read in full, re-derived	Verified
Section 13 correlation and symmetry join	29025–29449	Not read	Not audited
Same-step (increment) linearization	29450–29857	Read in full, re-derived	Verified
Normalized local converse	29858–30051	Read in full, re-derived	Verified, given the published cube constraint
Uniform selection of lower tests	30052–30064	Read in full	Followed, not re-derived

What I verified

- **Assembly.** The finite-group inequality $E_h\|a(\cdot+h)b\| \leqslant \|a\|\cdot\|b\|$ (with the stated powers of the Gowers norms) is correct. The two downsets $A = \{(0, j) : j \leqslant s - 1\}$ and $B = \{(i, j) : j \leqslant s - 2, i + j \leqslant s\}$ have union J, and a sequence supported on A is independent of h. The splitting lemma is correct: the product map into the two quotient nilmanifolds is injective because the lattice is exactly the set of points with integer Hall coordinates, which the integral coordinate lemma proves with group commutators as coordinates. The induction step and the passage from the limit statement to a finite family are the standard arguments and hold.

- **GI(2) base.** The box lemma (vanishing third differences on a box give a quadratic polynomial in the coordinates) is correct. The bracket identities are exact, the Heisenberg test function is well defined on the quotient, the sequence (βn, αn, 0) is a polynomial sequence for the degree-two filtration, the cutoff count 1 + 4d and the dimension 2d + 9d² are right, and the transfer from a prime cyclic group to the interval loses only fixed constants.
- **Tautification.** A polynomial nilsequence on a filtered nilmanifold with $G_0 \neq G_1$ equals one on the subgroup generated by the first positive groups, with a taut filtration, the same positive-degree groups and the same vertical character. The shift theorem follows: $\chi(n + u) \otimes \bar{\chi}(n)$ has degree support strictly below that of χ, on geometry that does not depend on u.
- **Lemma 13.2.** The replacement filtration on G^3, generated by the images (u, u, u), (u, u, 1) and (u, 1, u) by total weight, satisfies the commutator law. The triple (h(x + x′, y), h(x, y), h(x′, y)) is an exact ordered product of polynomial factors for it, its top group is {(uv, u, v)}, and the test function is invariant under that group. So the symbol is additive in each variable.
- **Difference decomposition and additive quadruples.** The decomposition of $\chi(h + k, n) \otimes \bar{\chi}(h, n)$ into a nilcharacter in n times a lower factor follows from the shift theorem, a splitting with an explicit lattice, and the published vertical Fourier lemma. The two Cauchy–Schwarz steps then give the modified Proposition 7.3. This is the route the authors' erratum describes for replacing the false Proposition 8.3.
- **Plain additive quadruples.** Both Cauchy–Schwarz steps, the change of variables with its prefactor, the count $(\eta/2B^4)(4N + 1)(2N + 1)^2$ and the injectivity of the quadruple map are correct. The output has the form of the published Proposition 8.4.
- **Converse.** Given the cube constraint of Green and Tao, the argument is complete: uniform tensor approximation, the normalized smooth average with $N \leqslant S \leqslant 4N$, the summable Fourier bound from two summations by parts, the embedding in a cyclic group without wrap-around, and the Gowers–Cauchy–Schwarz step.
- **Increment linearization.** The lattice lemma, the group $P_1 \rtimes \mathbb{R}$ with its filtration (which gives step at most s), the orbit identity $\pi(a^n) = h(n)\Gamma$, and the Lipschitz extension of the test function are correct. So a polynomial nilsequence of degree s is a linear nilsequence on a fixed nilmanifold of step at most s, with a fixed Lipschitz loss.

Claims about the published sources, checked against the sources

1. **The U³ paper.** The manuscript says Proposition 3.2 of Green and Tao (2008) is false and that only its use with a proper progression and trivial subgroup is needed. The authors' own note on arXiv:math/0503014 (version 4, December 2023) says the same. The statement of the theorem the manuscript imports (10.9 in the arXiv numbering, 10.11 in the journal) matches what the manuscript quotes, including "for any non-empty set $A \subseteq P$ there exists an h".
2. **Section 7 of Green–Tao–Ziegler.** The manuscript says the printed deduction on p. 1272 has the two coordinates of the multidegree sets interchanged. The arXiv text confirms it: the printed sets are $\{(i, 0) : i \leqslant s - 1\}$, called "independent of h", and $\{(i, j) : i \leqslant s - 2, i + j \leqslant s\}$. They do not cover J. The manuscript's sets are the intended ones.
3. **Appendix C of Green–Tao–Ziegler.** The printed proof of Proposition C.2 uses the full polynomial group with the shift and calls the result s-step. The manuscript says this fails. It does: for $s = 1$ and $G = \mathbb{R}$ the group is the three-dimensional Heisenberg group, which has step 2. The manuscript's increment construction avoids this.
4. **Acknowledged errata.** The authors' six-page erratum (2024) lists non-taut filtrations, the incorrect Proposition 8.3, the definition of the group G^* in Section 12 and a missing factorial in (E.3). The arXiv listing for version 5 (April 2026) says it corrects the proof of Lemma 13.2, where the wrong filtration was specified. The manuscript's replacement sections correspond to these items. It states that it uses the erratum only to locate the problems.

Not checked against sources: the statement of the cube constraint (Green–Tao, Linear equations in primes, Proposition 11.5) and the manuscript's remark on the proof of Proposition 11.2 there; the misprints it reports in Example 12.3 of the U³ paper and Example 6.11 of Green–Tao–Ziegler; the Hall-basis reference to Tointon's book; and every page-level claim about Sections 9–12.

Open points

- **The unaudited sections.** They are the frequency analysis and linearization step (Sections 9–12 of the published proof, with the factorization lemmas, the petal quotient and the rank iteration) and the assembly of the symmetry step (Section 13 with the factorial correction). This is the longest and hardest part of the argument.
- **The part describes itself inconsistently.** Its first sentence says the modules prove the theorem. Several units say otherwise about themselves: "this packet does not accept … pending Sections 9–11 argument or any downstream assembled inverse theorem" (lines 26074–26075); "not a declaration that all historical versions or the

entire inverse theorem have been checked" (27134–27135); "Acceptance of the finite iteration alone does not establish those analytic premises or the full inverse theorem" (28405–28406). The bibliography lists three internal units as awaiting independent acceptance or review (lines 39763–39768, 39843–39847, 39854–39860). I matched the eight interfaces of the assembly to the sections that supply them. I did not check that every such disclaimer is discharged somewhere else in the part.

- **References a reader cannot resolve.** The text points to working files and their line numbers, for example "Section 13 lines 327–343", "DF002 lines 132–145", `lp04-001`, `rank-iteration-001`, "the separate finding packet" and "the accompanying contracts". None of these is in the manuscript.

Cite or prove

The statement is in the literature, in three places: Green–Tao–Ziegler (2012) with its erratum and corrected arXiv version; Manners (2018); and Leng, Sah and Sawhney, arXiv:2402.17994. The erratum credits Leng, Sah and Sawhney with finding the errors and says their paper fixes all of them. Their Theorem 1.2 is finitary, requires $G_0 = G_1$, and gives a polynomial nilsequence on a nilmanifold of degree s with bounded dimension, complexity and Lipschitz constant. Its arXiv page showed no journal reference when I fetched it.

The one place where citation alone is not enough is the passage from polynomial to linear orbits of the same step, because the printed proof of Proposition C.2 gives step s + 1. The manuscript's increment-linearization section closes that gap, and I verified it in full. So GI(20) and GI(39) can be treated as an external dependency: cite the polynomial form, note in a short paragraph why bounded dimension and complexity give a finite family, and keep the increment section. On that route the other sixteen sections carry no weight for the main theorems.

If the manuscript instead rests on the internal proof, the eight unaudited sections and the rest of the petal-quotient section need a review by someone who works with the Green–Tao–Ziegler argument. This audit does not supply that.

Item 3: completeness and pole realization, reconstructed

Result. I reconstructed the completeness half and the pole-realization lemma step by step and found no gap. The reconstruction stops at three inputs that I did not re-prove here: the restricted converse, Kolyvagin's theorem with modularity, and the

simultaneous-primes theorem. The baseline's point charts, which the manuscript takes from an unprinted certificate, hold at every sample configuration I tried in exact arithmetic.

Pole realization. The predicate used in the 222-variable family is the zero-preparation case: for $b \neq 0$ and $B = 17b$, rational e, p, q, s with $p^2 = B(1 - e)(1 + 3e)$ and $B \cdot q^2 = (s^2 - Bh)(s^2 + 3Bh)$, where $h = e(B - b^2)$.

- *Soundness* (a solution excludes a negative odd valuation of b outside 2, 3, 17). Re-derived: the valuation of e is forced to be zero, then Bh has odd valuation, and the right side of the second equation has even valuation while the left has odd valuation or is zero. The degenerate witness values are covered.
- *Completeness* (a solution exists when $0 < b < 17$, the reduced denominator of b is a square, b is a square in $\mathbb{Q}_2$, $\mathbb{Q}_3$, $\mathbb{Q}_{17}$, and $v_2(b) \geqslant 3$, $v_3(b) \geqslant 1$, $v_{17}(b) \geqslant 2$). This reduces to finding one squarefree n for which the two curves E_n and E_{nθ}, where E_ℓ is $y^2 = x(x - \ell)(x + 3\ell)$, both have a rational point whose abscissa lies in a prescribed square class β. I re-derived the whole argument: the local images of the 2-descent at primes dividing ℓ, at 3, at 2 and at infinity; the three linear conditions over $\mathbb{F}_2$ that a Selmer class must satisfy; the identity $K + K^T = ss^T + \mathrm{diag}(s)$; the choice of auxiliary primes ($d - 1$ primes congruent to 1 and one prime congruent to 7 modulo 12) and of their residue symbols; the nonsingularity of the deleted matrix, including the multilinear argument for the free bits; the conclusions $\ker K = \langle 1, 1_\beta \rangle$ and $\lambda \notin \mathrm{im}\, K$; and hence a 2-Selmer group of dimension exactly three on both curves, with the class (β, 1) independent of torsion.
- Two steps are correct but not spelled out in the text. The kernel of $K + \mathrm{diag}(s)$ is a line because s is not in the image of K, which follows from $K^T \cdot 1 = 0$ and $s^T \cdot 1 = 1$. And the class (β, 1) comes from a point of infinite order because the independence of the two torsion classes rules out rational 4-torsion.
- From dimension three to a point: the corank lemma (Cassels–Tate pairing) gives 2-primary Selmer corank one; the restricted converse gives analytic rank one; Kolyvagin's theorem gives rank one and finite Ш; then the Kummer sequence turns the Selmer class into a rational point. The specialization to preparation zero ($V = B^2(1 - b/17)$, with the two squarefree parts congruent to 17 and 1 modulo 24 and mutually square) checks.

Completeness of the 248-variable baseline. For an integer input, in the order the manuscript makes its choices:

- *Scalar pole.* On fixed real neighbourhoods the ratio Z is bounded and |Norm f(H)| ⩽ 1/7, so the scalar pole input is positive and tends to 0 as N grows, uniformly in the input. Its square class is that of R(R + 9792), whose reduced denominator is a square because numerator and denominator are coprime, and which is a square at 2, 3 and 17 by the valuations of R. Choosing the reference point deep at the finite support and the comparison index deeper makes the remaining factor integral where needed. All of this I checked, including the congruences behind the valuations of R.
- *Real and congruence conditions together.* The density lemma for progressions in the closure of the orbit holds as stated: if L is divisible by the order of the component group, a progression modulo a multiple of L is dense in the component of the reference point.
- *The two detectors.* The contact quantity equals the product of the ten selected primes times a rational square, so its valuation is even away from those primes and equal to 1 at them. At a selected prime, which splits in K, the congruences h ≡ 0 and h ≡ 1 modulo the two reduction orders make H and the U_i formal at one place and H − P and the V_i formal at the conjugate place; both logarithmic ratios equal the integer label, so the comparison tests vanish to order four and every weighted term has positive valuation. At the prepared primes the weight tends to 0, the contact quantity to 1, and the pole input acquires the valuations 4, 2, 2 at 2, 3, 17 from the fixed factor 408/409. Over the reals it tends to $(204/409)^2$, which lies in (0, 17). So each detector's pole input meets the hypotheses of the pole lemma.
- *Labels.* The Lucas construction of the root labels works for every a ⩾ 3.

The later arithmetic changes. The same reconstruction goes through after each of the three changes. For 248 → 245 the scalar ratio becomes z(S)/z(H), which is what the new row says. For 238 → 237 the new pole variable is $O(R^{-4})$, the pole input is $O(R^{-6})$, and the stated witnesses $C = u^2/R$, $D_1 = v^2$, $w = R^2W$ satisfy the two changed rows identically. For 233 → 232 the valuation of R at 3 changes from 1 to 0 and the local square condition still holds, as checked in the first pass.

The baseline charts. The manuscript states that the fixed-subtraction formulas and the root charts encode the group law, and refers to a certificate in the repository for the proof. I checked them directly with actual points, in exact arithmetic over $\mathbb{Q}(\sqrt{2})$: the formula for z(S − P) on 54 sample charts and the formulas for f(H − P) and z(H − P) on 54 sample points; and, for 55 configurations Q = [n]P, U = [u]P, the ten block equations hold with V = U − Q for exactly two of the four sign choices and never with V = U + Q. This confirms that genuine point configurations give witnesses. It is a check at sample points, not a symbolic proof.

One small discrepancy. The text says the pole hypotheses are verified "for each of the four normalized pole inputs". The tables have three pole systems: two detectors and the scalar.

What the completeness half rests on. The restricted converse for the curves E_ℓ with $\ell \equiv 11$ modulo 12 (2-primary Selmer corank one implies analytic rank one), which is the theorem that uses Kato's Euler system; Kolyvagin's theorem and modularity; and the simultaneous-primes theorem for Gaussian norm forms, which now also rests on the manuscript's internal inverse theorem. None of these was re-audited in this item.

Script: `/home/claude/audit/v15/chart_check.py`.

Item 4: the reduction chain from the 248-variable baseline, by machine

Result. The chain is now machine-checked from the stored baseline table to the printed 222-variable quartic. For the part that was open (248 → 245 → 240 → 239 → 238 → 237 → 235 → 233), every printed system and every stored table agrees with what the manuscript says each step does, and each two-sided step rests on exact polynomial identities that I verified with my own code. Three steps change the arithmetic construction and are not equivalences: 248 → 245, 238 → 237 and 233 → 232. For those the machine check covers the change of rows only; their arithmetic belongs to Item 3.

What data exists. The repository stores the 248-variable baseline (`family248_v17/quadratic-family248-v17.json`, 222 rows) and tables for 237, 235 and 233 variables. It stores no table for 245, 240, 239 or the raw 238 system. The manuscript describes each step, gives a coordinate dictionary for the raw 239 system, and prints the 238 system. I therefore rebuilt the passage 248 → 238 from the manuscript's description alone and compared the outcome with the printed 238 system.

248 → 238. I wrote every coordinate of the 248 table as an explicit expression in the coordinates of the raw 238 system, using only what the manuscript states: the dictionary, the table of R-weights, $\rho = 1/\lambda$, and the inverse of the root-product map. Then:

- Of the 137 retained rows outside the four root blocks, 132 land in the span of the printed 238 equations with no extra factor, and 5 become identities (the three old power definitions attached to λ, and the definitions of R^2 and R^4).
- The 72 component rows of the nine new root-product equations for the four blocks, the two rows of the formula for j_S, and the two rows of the new scalar ratio $Z \cdot j_S = h_S \cdot S_H$ also lie in that span.

- Together these have rank 208, which is the rank of the printed system modulo its four definitional rows (R^2 and the three powers of ρ). No printed equation is left over: 132 + 72 + 2 + 2 + 4 = 212.
- The 80 block rows of the 248 table are the components of the ten field equations per block that the manuscript lists. The equivalence between those ten and the nine new ones holds by identities that I verified in exact arithmetic over $\mathbb{Q}(\sqrt{2})$, in both directions. In the inverse direction each old equation becomes a combination of the new ones with polynomial multipliers; in the forward direction each new equation, multiplied by a power of h, is a combination of the old ones. The only quantities inverted are h and $x = h^2$, which are nonzero because $j^2 = h^4 - D$ has no solution with $h = 0$ at a real embedding.
- For the other divisions: $R > 0$ because the contact product is a polynomial in X^2 with positive coefficients; $\rho \neq 0$ because $\rho \cdot t_1$ equals a positive trace; and $\lambda \neq 0$ in the old system because the contact polynomial vanishes at 0, which I confirmed.

One convention is not stated in the manuscript. In the 248 table the chart of the third root stores half of the real component of x. With the uniform convention, my first run flagged the trace row as inconsistent (weights 1, 1, 2, 1 against 1, 1, 1, 1). With the table's own convention, read from its row 9, the rows agree. This concerns the stored baseline table only, not the printed systems.

238 → 237 → 235 → 233.

- Printed 238 against the 237 table: 209 of the 211 rows of the 237 table, moved back to raw-238 indices and through the eight printed coordinate stages, lie in the span of the printed 238 equations. The other two are exactly the manuscript's two new rows. Those 209, together with the three old rows reconstructed from the dictionary, span the printed system (rank 212).
- 237 → 235: six rows are the components of $\chi f = 1$, $S^2 = \chi - Df$ and the old reference equation, and χ occurs nowhere else. The 235 table is the other 205 rows plus the components of the two new field equations. Both printed identities hold.
- 235 → 233: four rows are the components of $g^2 = fv$ and $w = v^2$, and v occurs nowhere else. The raw 233 table is the two components of the square equation followed by the 205 retained rows in the stated order. The identities used in the two-sided argument hold.
- The raw 233 table is byte-identical to the starting table of the 222 package, so this joins the chain verified in the first pass.

What this does not cover. The three arithmetic changes named above; the 149-adic facts that exclude the vanishing of a + cf in the last two steps, which come from the scalar criterion; and the derivation of the 248 table itself from the construction credited to José Brox.

Scripts, in `/home/claude/audit/v15/`: `back_248_238.py`, `block_identities.py`, `back_238_237.py`, `back_237_233.py`, with the shared `texparse.py` and `klib.py`.

Item 5: the arithmetic behind the 314-variable quartic

The 314-variable polynomial ends a chain of effective constructions that is separate from the 222-variable family: Version 9 (523 and 540 variables), Version 12 (498), Version 13 (394 and 411) and Version 14 (299 and 314). Each version changes the arithmetic certificate, so each has its own soundness and completeness argument in the manuscript (lines 20352–23764). The first pass confirmed only that the printed table is the sum of squares of its 294 rows. In this pass I read the four sections in full, re-derived each argument, and compared the stored 314-variable system with the equations the text argues about.

Result. I found no error. Every inference made inside these four sections holds, and at every place I compared them the stored system is the system the text describes. The theorem still depends on results proved earlier in the manuscript; the last part of this section lists them.

Versions 9 and 12 (lines 20352–21947)

Verified, by re-derivation unless a script is named:

- **Logarithm truncations.** The recursion $a_0 = 1$, $a_1 = -2/5$, $a_j = (4j - 5)/(4j + 1)\cdot a_{j-1}$ reproduces the series of $\log(Q)/z(Q)$ in $w = D\cdot f^2$ (checked by machine through degree 14). So the truncations after degree 2 and degree 9 have relative errors of order 12 and 40 at depth one. The mixed-precision lemma (lem:v12-mixed-local) follows: one of h, h − 1 is a unit, the cross-test transfers integrality, and the labels are recovered to orders 12 and 40.
- **Anchor.** For even $s \neq 0$, the ρ-coefficient f_1 of $f([s]P)$ has 3-adic valuation $2 + 2v_3(s)$ (lem:v12-anchor-coefficient). I re-derived this from the 3-adic formal expansion and confirmed it by exact computation for s up to 40. The anchor equation $u^2 + 6v^2 - 2w^2 - 4Cw - 3f_1^2 = 0$ has a rational solution exactly when f_1/C is integral at 2 and 3. The manuscript takes this criterion from a quaternion-algebra statement in Voight's book.

I re-derived it independently with Hilbert symbols: the form ⟨1, 6, −2⟩ is anisotropic at 2 and at 3 and misses exactly the square class of 3. With C = 3^(2²⁵) the anchor forces $v_3(s) \geqslant 2^{24} - 1$, hence $\log_2|s| > 25165823.5$.

- **Scalar norm tree.** The valuation chain in lem:v12-scalar-sound gives $v(z_0)$, $v(z_1) \geqslant 32$ at every prime where the contact product has odd valuation. The duplication step handles a point near (0, 0). The conclusion is that the 32nd power of the contact radical divides s.
- **Reciprocal chart for S − P.** The five polynomial identities hold (`recip_chart.py`). On the equal-abscissa branch the formulas give the point −2P, whose coefficient has 3-adic valuation 2, so the anchor excludes it as the text says.
- **Exact private changes.** The weight circuit ($q_{12} = q \cdot e^{22}$, $q_{40} = q \cdot e^{78}$), the detector pole normalization and the removal of the scalar norm coordinate are inverse changes of variables.
- **The two height contradictions.** Each numerical step holds: $|s| > (80\text{DONT PANIC})^{10}$; $|F| < 5B^4$ for B > 4; B^(1/10) < 7DONT PANIC·5^(19/40) + 6 < 24DONT PANIC; B^(5/12) < 40DONT PANIC·e^((19/12)h(β)); log B < (31/200)|a| < |a| − 3.
- **Completeness.** The order of choices is consistent, and the Lucas labels are odd integers: $U_j \equiv j$ modulo a − 2, the sequence has period 3 modulo 2, and k = 2(a − 2) ≡ 1 modulo 3.

Version 13 (lines 21948–22810)

Verified:

- **Rational logarithm.** $Q \cdot G_9 - N$ equals the two printed terms in w^{10} and w^{11}. The nine Bernstein coefficients of N are as printed and positive, and Q decreases on [0, 1] to 60/1073. N has no root modulo 71 and $Q \equiv 1 + w^2$ has none modulo 7; both primes split in K, so neither polynomial vanishes anywhere on K (`v13_checks.py`, `v13_biquad.py`).
- **Charts.** The odd-root chart ($A^2 - 2(x + j)A - D = 0$ with Norm(A) = $-u^2$) selects a half-point in the right square class. The identity-inclusive comparison chart and its scaled biquadratic form satisfy their identities (`v13_chart.py`, `v13_biquad.py`), and the boundary cases r = 0 and g = 0 give −Q and the identity.
- **Larger domain for H.** The three norms ($41^2/64$, $17^2/64$, $63^2/64$) and the three translated abscissae are correct. So f(H) is a unit at every prime above 76176 when H is a torsion point or lies in ±P + E[2], and the height bound excludes these points. The group identities $[m]U_i = [2u_i]H$, $[m - 2]V_i = [2(u_i - n_i)]V_0$ and m = [4s]H replace the old half-indices correctly.

- **New label and correction.** Put $\tau = 2X + 1$, $\beta = 51\tau^2 + 2504$ and $c = r + 4896$. For integral X, $v_2(r) = 2$ with $r/4 \equiv 1$ modulo 8, $v_3(r) = 1$, and $r \equiv 15$ modulo 17, which is a square. So $c \cdot r$ is a square in $\mathbb{Q}_2$, $\mathbb{Q}_3$ and $\mathbb{Q}_{17}$ (by hand, and by machine for $-300 \leqslant X < 300$). An odd integral β forces X to be integral; the text's warning that $X = -1/2$ gives the even value 2504 is correct.
- **Depth 24.** The chain $v(B_s) > 63\eta/4$, $v(q) > 61\eta/4$, $v(z_0) > 191\eta/8$, $v(z_1) > 187\eta/8$ gives valuations of at least 24. Hence $h(\beta) \leqslant$ 🪨·|s|^(19/24), and (5/24)·25165823.5 exceeds 2093973 as required.
- **Second height contradiction at order 10.** B^(1/10) < 48🪨·e^((19/10)h(β)) and log B < 29|a|/40 < |a| − 3.
- **Exact private changes.** The new weight circuit ($q_{10} = q \cdot E^{18}$, $q_{40} = q \cdot E^{78}$), the factored pole conic with its zero lift, the preparation conic in both directions, the pole normalization by powers of x and the anchor rescaling identity all hold.
- **Preparation set to zero.** The specialization needs $0 <$ 🪨 < 17, a square reduced denominator, and local squares at 2, 3, 17 with valuations at least 3, 1, 2. The construction supplies $v_2(cr) = 4$, $v_3(cr) = 2$ and a 17-adic valuation of at least 8. The order of choices (scalar point, then H in a real neighbourhood, then contact primes) is consistent.

Version 14 (lines 22811–23764)

Verified:

- **H chart.** The certificate identity for the fixed-difference chart and the three identities of its squared form hold (`v14_checks.py`). The two exceptional branches are as stated: U = 0, and Q = 0 with $v = (3 + 2\rho)/4$ or $(-36 + 16\rho)/49$.
- **Root parity from the shifted anchor.** Write x_n for the ρ-coefficient of x([n]P). The valuation table ($v_3(x_n) = 0$ for odd n and $-2 - 2v_3(n)$ for even n; $v_2(x_n) = -1$ for odd n) is correct: re-derived, and exact for $n \leqslant 40$ in `v14_valuations.py`. The shifted anchor value is $T/C = F_s/C + 12(x^2 + x'^2) + 108 \cdot x''^2$, with x, x′, x″ the coefficients for the roots A, T, Z. Its three summands have 3-adic valuations in $2\mathbb{Z}$, $1 + 4\mathbb{Z}$ and $3 + 4\mathbb{Z}$, so they cannot cancel. Integrality therefore makes the three labels odd and s even, and restores the old anchor.
- **Common-scale groups.** For every odd α the valuation classes modulo 16 of the terms of G_0, G_1, G_2 are pairwise disjoint and avoid 0 (machine). The consequences follow: each group has a unique minimal term; at one place every term has valuation above $-\alpha/2$; $v(\gamma) > 15\alpha/32$; $v(H_i) \geqslant 40$ and $v(M_i) \geqslant 10$; the reciprocal depths are 40, 20 and 22.

- **Reduced reciprocal charts.** The slope, alternate-cubic and root-curve identities hold (machine). The orientation and the branches f = F, U = O and V = O are handled correctly.
- **H-depth bootstrap.** The Lucas polynomial is −1 at the origin, so one of L_2, L_3, L_4 is a unit. That gives depth at least 10 for H, and the orders 40, 12 and 10 follow in that order without circularity.
- **Completeness.** The valuation bounds at contact primes (1, 2, 9, 10, 159, 3, 83, 76, 8) and the table at 2, 3, 17 are correct. Tying $\gamma = 1/u$ to the anchor quadric $u^2 + 6v^2 - 2z^2 = 3T^2 - 2C^2$ works: the stated local points at 2, 3, 17, at contact primes and at the real place solve the quadric, and the reflection map Φ gives weak approximation.
- **Constant chain.** The exponents (2049, eleven times 2048, 0, 0) give $3^{(2^{25})}$.

The stored system against the text

I compared rows of `candidate314-ir.json` with the manuscript's equations by exact polynomial arithmetic.

Statement in the text	Rows	Outcome
Scalar contact product $r(\beta)$, $\beta = 204(X^2 + X) + 2555$	453–455, 512, 566	The coefficient polynomial is $-r(\beta)$ exactly
$(r^2 + 4896r)\cdot V + 82479248\cdot b_0 = 0$	492, 560	Exact
Comparison test $M_1 = L_1 - \beta(X)$	259, 260, 910, 911	$ZM_1\cdot\gamma^{40} = (L_1 - \beta)^4$ exactly
Contact value $R = F_5(\lambda)$	298–302	Coefficients agree
Groups G_0, G_1, G_2 for detector 0	919–924, 2010–2013	Agree term by term, including conjugation
Squared H chart	34, 35, 112, 113, 130, 131, 661, 662	Exact
Scalar reciprocal equation and its guard	513, 514, 732	Exact
Primary ratio with the fixed normalizer	184, 185	Exact
Shifted anchor, T and Q (weights 48, 48, 432)	504, 1903, 1000	Exact

Statement in the text	Rows	Outcome
Scale chain γ^{16}, γ^{40}, γ^{80}, γ^{160} from the anchor coordinate u	2000–2007	Exact
Constant chain	1001–1014	3^{2049} and 3^{2048} exact

One thing here looks like a discrepancy and is not. The coordinate named `testSquare.12.1` stores $(L_1 - \beta + 3)^2$, and the consuming rows 910 and 911 apply the shift. The tested quantity is $L_1 - \beta$, as the text says. The 498-variable system uses the same convention.

Not compared: the groups of detectors 1 and 2, which have the same row pattern with a different scale, and the remaining rows of the system.

What the 314-variable theorem rests on

Versions 9–14 use the following results, which are proved earlier in the manuscript. I did not re-derive them in this pass.

- **The height estimate** $h(a) \leqslant H_{19} \cdot M_5(a)^{19}$ with $\log_2 H_{19} < 2093966$ (thm:quartic-height19), and the root-height implication $B^2 - 6 < 7\text{[illegible]}\cdot|N|^{(19/k)}$ built on it. Every numerical margin above depends on this constant. Its proof uses the Faltings-height results whose statements item 6 checked in part.
- **The pole obstruction and pole realization lemmas.** Item 3 re-derived both. Realization depends on the restricted converse (item 1), on Kolyvagin's theorem with modularity, and on the simultaneous-primes theorem.
- **The simultaneous contact-prime theorem** (thm:v5-simultaneous-primes) with its real-density condition, used directly in every completeness argument. Its proof (lines 17544–20351) uses the Green–Tao relative inverse theorem and so depends on the inverse theorem of item 2.
- **The preparation-zero lemma, the Lucas lemma, the formal-logarithm estimates** (Silverman IV.6.4 and the two reciprocal-logarithm lemmas), and the Mordell–Weil facts $E(K) = \mathbb{Z}R_0 + E2$ and $2E(K) = \mathbb{Z}P$.

For Versions 9, 12 and 13 the text also says that the earlier pole-completeness estimates still hold for the changed test orders, and quotes their decay rates from the Version 5 source. I followed this and did not re-derive it. Version 14 replaces it with the explicit argument checked above, so the 314-variable theorem does not depend on it.

Two wording slips, neither affecting the argument. In the Version 13 exclusion the text says “m is neither 0 nor 2” where the argument excludes 0 and ±2. The stored coordinate names keep “12” for tests that the Version 13 and 14 text calls order 10.

Item 6: cited theorem applications against their sources

I fetched each source that was reachable and compared the cited statement, with its hypotheses, to the way the manuscript uses it.

Citation in the manuscript	Source consulted	Outcome
Javanpeykar, Prop. 4.1.3: bound on the different (Lenstra)	Published article (Algebra & Number Theory 8)	Matches exactly. The proposition reads r_β ⩽ e_β − 1 + e_β·ord_A(p^m), with m the largest integer such that p^m ⩽ n. With ord_A(p) = e_[redacted] and n = d this is the manuscript′s inequality. The paper′s Remark 4.1.2 covers imperfect residue fields, as the manuscript says
Javanpeykar, Lemma 2.3.1	Same	The pointer is slightly off. The lemma′s statement is a height inequality; the blow-up formula for the canonical sheaf, which is what the manuscript uses, is in its proof. The fact itself is standard
Gaudron–Rémond 2023, Thm 1.9(1)	Published memoir (Mém. SMF 176)	Matches. Isogenies in both directions of degree at most Ξ(A)², with Ξ(A) = ((7g)^(8g²)·[K:ℚ]·max(1, log[K:ℚ], h_F(A)))^(2g²), stable Faltings height in Faltings′ original normalization. For g = 2 this is the manuscript′s (14³²·d′·max(…))¹⁶

Citation in the manuscript	Source consulted	Outcome
Gaudron–Rémond 2023, Lemma 16.2(1), pp. 97–98	Same	The manuscript uses the construction in the proof, and describes it accurately: an isogenous variety with maximal multiplication and no larger height (Rémond 2022, Cor. 1.3), which is a product of simple abelian varieties (Rémond 2017, Thm 1.6), then Bost's inequality. I did not check the two Rémond results
Modularity at p = 5, as quoted from Calegari's ICM survey, p. 624	arXiv version of the survey, §7.2	Matches the survey's sentence: Pan proves modularity up to twist of any geometric ρ: $G_{\mathbb{Q}} \to GL_2$ for $p \geqslant 5$, assuming distinct Hodge–Tate weights and oddness. This is a survey sentence; I did not check Pan's theorem in his paper
Carayol 1987, §§2.1, 2.4, 4.1: conductor of the ℓ-adic representation	Numdam scan	Matches. Théorème 4 is compatibility with the local correspondence at every prime other than ℓ, including the primes of the level; the conductor equality is stated as its consequence in §2.4
Dokchitser–Dokchitser, Thm 1.4	arXiv version	Matches: the p-parity conjecture for every elliptic curve over $\mathbb{Q}$ and every prime p, including 2
Colliot-Thélène–Salberger 1989, Prop. 1.3	Author's posted copy of the article	Matches: over a field of characteristic zero, a geometrically integral cubic hypersurface that is not a cone and has a rational point is unirational; the normal case is by reference to Colliot-Thélène–Sansuc–Swinnerton-Dyer, Remark 2.3.1, as the manuscript says

Citation in the manuscript	Source consulted	Outcome
Von Buhren 2017, Théorème 1.1	Numdam (first pass)	Matches
Kato 2004	Numdam scan	See Item 1
Kato 1999	Not reachable	Not checked; see Item 1
Ribet 1990, Prop. 3.3, Remark 3.5, Prop. 3.6	Author's posted copy	Not checkable here: the file is a scan without a text layer

Not attempted: the two Rémond papers, Gaudron–Rémond 2014 (the normalization and Bost's lower bound, which the 2023 memoir also uses), von Känel 2021, Pan's article, the Stacks Project tags, and Colliot-Thélène–Sansuc–Swinnerton-Dyer's Remark 2.3.1. None of these is used in a way that looked unusual, but I have not read them against the text.

No citation that I could check is misapplied.

Findings for the second pass

No error found. In everything examined in this pass, every inference holds and every finite claim reproduces. That covers the reduction chain from 248 variables, completeness and pole realization, the four effective constructions ending at 314 variables, the repair steps of the Kato application, and the audited sections of the inverse theorem.

What remains unvalidated

In order of weight:

1. **Kato's 1999 paper.** The repaired application at p = 2 uses a reading of sections 0.1 and 0.6–0.8 of that paper. I could not obtain it, so that reading is unchecked. Separately, parts (1) and (2) of Theorem 13.4 in the Astérisque paper carry no restriction to odd p, but the paper gives no proof of them and attributes them to three other papers. If you attach the 1999 paper I can finish this item.
2. **The inverse theorem.** GI(k) enters the completeness arguments through the Green–Tao prime asymptotics used in the prime-selection theorems (lines 2217, 9565, 19474–19476 and 19704). This includes the simultaneous-primes theorem behind the 314-

variable quartic. The internal proof is verified only in part. The statement is published, so this dependency can be closed by citation together with the verified increment-linearization section, as set out under item 2.

3. **Inputs outside the six items.** The height estimate with the constant H_{19}, Kolyvagin's theorem with modularity, the simultaneous-primes theorem, and the formal-group and Mordell–Weil facts were not re-audited in this pass.
4. **Citations.** Ribet 1990 and Kato 1999 could not be read, and seven further citations were not attempted. They are listed under item 6.

Corrections to the text

None of these affects a proof.

- Line 32294 says "four normalized pole inputs". The tables have three.
- The 248-variable table stores half the real component of x in its T chart, and the text does not say so. My first run of the chain check failed on exactly this.
- The pointer to Javanpeykar's Lemma 2.3.1 should be to its proof.
- In Version 13, "m is neither 0 nor 2" should exclude 0 and ±2, as the argument does.
- Stored coordinate names keep "12" for tests the text now calls order 10, and `testSquare.12.1` stores $(L_1 - \beta + 3)^2$ with the shift applied by its consumers. Both are naming matters; the tested quantity is the one in the text.
- The inverse-theorem part refers to working files and line numbers that are not in the manuscript, and several of its units say they are awaiting review while its opening sentence says they prove the theorem.

Claims about published sources that I confirmed

- Proposition 3.2 of the Green–Tao U^3 paper is incorrect, by the authors' own note of December 2023, and the case the manuscript needs survives.
- The deduction on p. 1272 of Green–Tao–Ziegler prints its two multidegree sets with the coordinates interchanged.
- The printed proof of Proposition C.2 of Green–Tao–Ziegler gives a nilmanifold of step s + 1, not s.
- The authors' erratum and their corrected arXiv version acknowledge the other defects that the manuscript's replacement sections address.

What I would do next

- Check Kato's 1999 paper, if you can attach it.
- Decide whether the inverse theorem is cited or proved internally. If it is proved internally, the eight unaudited sections need a specialist review.
- If you want the dependency chain closed end to end, the height estimate and the simultaneous-primes theorem are the two largest inputs that no pass has re-derived.

Third pass on Version 15: four further items (10 Nov)

Scope of the third pass

Astra asked for four further items on the same commit (`511529b`, manuscript `quartic_v15.tex`). Each has its own section below.

Item requested	What this pass did	Outcome
The remaining inverse-theorem sections	Read all eight in full and re-derived their arguments	No error. One missing normalization, with its repair.
Kato 1999, Ribet 1990 and the other unchecked sources	Checked the listed sources I could reach, and further sources in the two dependency chains	Kato 1999 and Ribet 1990 are still unread. Those I checked match, with one citation slip. A few minor ones remain unchecked and are listed.
The height-estimate and simultaneous-primes dependency chains	Read both in full, traced every link to its published input, recomputed all constants	No error. Both are conditional on named published theorems.
Symbolic verification of the baseline point charts	Proved every chart identity as a polynomial identity on the curve	All fourteen checks pass

Terms. *Verified* means I re-derived the step by hand or reproduced it with my own code. *Followed* means I read the argument and found no gap, without re-deriving it. *Not audited* means I did not check it. A result is *conditional* when it rests on something this audit did not validate; that is a statement about the audit and not about the proof.

Method. I did not run any code from the repository; the sandbox does not allow it. All computations use my own scripts, with exact arithmetic: `charts_symbolic.py` for the charts and `height_checks.py` for the height constants. Published sources were read on arXiv, on authors' pages and on publishers' open-access pages, and each source list says which version I saw.

What this pass did not do.

- It did not audit the proofs of published theorems. Where a published statement is used, I checked that the manuscript applies it within its hypotheses.
- It did not obtain Kato 1999 or Ribet 1990. Project Euclid refuses automated access to the first, and the only copy of the second I could reach is a scan without text.
- It did not re-audit inputs outside the four items: Kolyvagin's theorem with modularity, the formal-group and Mordell–Weil facts, and the application of the Green–Tao theorem at line 2217.

Symbolic verification of the baseline point charts

Result. The baseline charts are now proved as identities. In the second pass I had only confirmed them at exact sample points. All fourteen symbolic checks pass, and nothing in the text needs to change.

The manuscript does not print a proof of these charts. It refers to "the supplied certificate" in "the exact function field" (lines 32533 and 32691). I did not use that certificate.

Method. Each chart is a set of polynomial relations on a curve, or on a product of two curves. I clear denominators and check that the resulting polynomial is zero modulo the defining relations, in exact arithmetic over $K = \mathbb{Q}(\sqrt{2})$. No points are substituted. The script is `charts_symbolic.py`.

Chart	Statement	Outcome
S, with $x = h^2$ and $j^2 = x^2 - D$	$z(S - P) =$ `N_S` / `D_S`	Verified
S	$f(S - P) =$ `g_S`^{2} (not stated in the baseline, used later)	Verified
S	`D_S`$^2 = 1024 \cdot x(P) \cdot (x(P) \cdot x(S) + D)^2 \cdot (1 - D \cdot$ `g_S`$^4)$, so `D_S` $\neq 0$	Verified

Chart	Statement	Outcome
S	`g_S` = 0 only at S = P	Verified
H, with χf = 1 and `S_H`² = χ − Df	`F_0` = f(H − P)	Verified
H	z(H − P) = `N_H` / `D_H'`	Verified
H	`D_H'` = 0 exactly when D· `F_0`² = 1, that is, when H − P is a 2-torsion point (±d, 0)	Verified
H	`N_H` and `D_H'` are the S-chart constants times the unit 3 + 2ρ	Verified
Root block	The block defines points U and V on E from its coordinates	Verified
Root block	V = U − Q for the root Q, in both coordinates	Verified
Root block	The same identity with V = U + Q fails (control)	Verified
Root block	For any U of this form, the block has exactly one solution	Verified

The steps that are not polynomial identities, checked by hand:

- **`D_S` is never zero.** By the third row it could vanish only if D· `g_S`⁴ = 1. That would give d· `g_S`² = ±1 with d = 3 − ρ, and taking norms 7·N(`g_S`)² = 1, which is impossible. This is the text's claim "Then `D_S` ≠ 0".
- **The degenerate strata.** The text says the charts permit U = O or V = O. With the first block coordinate equal to zero the equations give U = O and V = −Q. With the second equal to zero they give V = O and U = Q. Both are consistent with V = U − Q.
- **The ratios.** Given the identities above, the three ratio equations of the baseline mean what the text says: Z = z(S − P)/z(H), L = z(U)/z(H) and the unconjugated alternate ratio z(V)/z(H − P).

Two remarks. The formula for the fixed-subtraction coordinate has denominator x(P)·χ + D, which vanishes only at the abscissa of ±(P + (0, 0)); the identities hold away from that point, and the text excludes it. The S and H charts are the same formulas as the reciprocal charts of Versions 12 and 13, which item 5 of the second pass verified in that notation.

Sources examined directly

Result. Kato 1999 and Ribet 1990 are still unread: neither could be opened from this session, and I did not go around the sites that refuse automated access. Of the other unchecked sources, nine are now compared with the source text and all agree with what the manuscript takes from them. One citation points to the wrong paper.

Source	What the manuscript takes from it	Outcome
Kato, Kodai Math. J. 22 (1999)	Sections 0.1 and 0.4–0.7, Example 0.6.2, Theorem 0.8 (lines 2782, 13524, 13782–13867)	**Not read.** The publisher's direct PDF link still returns a bot-protection page, and the second host refuses automated access
Ribet, Invent. Math. 100 (1990)	Propositions 3.3 and 3.6, Remarks 3.5(a) and (b) (lines 15105–15116)	**Not read.** Both posted PDFs are image scans with no text layer. The Göttingen digitization needs a page that requires your approval to open, and the request timed out
von Känel, Forum Math. Sigma 9 (2021)	Faltings normalization (§2.1), Bost's lower bound (2.2), degree and genus of congruence curves (4.1)–(4.2), $g \leqslant N^2/24$ (5.9)	Match, all six uses
Gaudron–Rémond, Polarisations et isogénies, Duke 163 (2014)	Theorem 1.4 (isogeny bound) and the bound $h \geqslant -3g/2$ in Section 9	Match
"Gaudron–Rémond 2014, §2.3 and Corollary 8.4" (lines 30633 and 30646)	Normalization of the Faltings height, change under isogeny, Bost's bound $-g\cdot\log(\pi\sqrt{2})$	The statements match, but they are in a different 2014 paper. See below
Rémond, Propriétés de la hauteur de Faltings (2022)	Corollary 1.3, used through Gaudron–Rémond 2023, Lemma 16.2	Match (author's preprint of June 2021; the journal version is behind a paywall)

Source	What the manuscript takes from it	Outcome
Pan, arXiv:1901.07166v2	Theorem 1.0.4	Match: p odd, with an extra condition only when p = 3
Green–Tao, Linear equations in primes	Proposition 11.5; the manuscript's remark on displays (11.6)–(11.7)	Match. The remark describes the printed proof accurately
Green–Tao–Ziegler, Example 6.11	Two misprints reported by the manuscript	Both are present in arXiv version 5
Stacks Project	Fourteen tags (listed below)	Match
Colliot-Thélène–Sansuc–Swinnerton-Dyer (1987), Remark 2.3.1	The normal case of cubic unirationality	**Not read directly.** Kollár's published description of that remark agrees with the manuscript's

The citation to correct. Lines 30633 and 30646 cite "§2.3" and "Corollary 8.4" of the bibliography entry for Gaudron–Rémond, *Polarisations et isogénies*. That paper has neither; it refers to them itself as belonging to Gaudron–Rémond, *Théorème des périodes et degrés minimaux d'isogénies*, Comment. Math. Helv. 89 (2014), 343–403. I read that paper: §2.3 fixes the normalization and gives the change of height under an isogeny, and Corollary 8.4 is Bost's lower bound. The mathematics is unaffected. The second paper should be added to the bibliography. The constant $-g\cdot\log(\pi\sqrt{2})$ is also stated in Rémond's paper and in von Känel's display (2.2).

Stacks tags checked. 00R4 (miracle flatness), 0AVT (reflexive modules), 0AWQ (dualizing module of a Cohen–Macaulay scheme), 0C0I and 0BUL (relative dualizing module of a finite or quasi-finite map), 0BWB (the different is the Kähler different), 03Q9 (stalks of higher direct images), 03QA (Leray), 03YR (stacks with trivial inertia are algebraic spaces), 0ADD, 0A18, 0D3A, 0D32, 0B45 (the chain from a proper algebraic space to a projective scheme). Each says what the manuscript uses it for. The manuscript prints the reflexive-module lemmas as 31.13.12–14; the site now numbers them 31.12.12–14 under the same tag.

Still not checked. Rémond's 2017 paper, Tointon's book, the misprint the manuscript reports in Example 12.3 of the U^3 paper, and tags 0E83 and 0BW0. None is a premise of a proof step I examined, except 0E83 (genus is locally constant in a smooth proper family).

What would close the two main items. Attach the PDFs of Kato 1999 and Ribet 1990 to this conversation. For Kato I need pages 314–318; for Ribet, pages 440–443.

The remaining inverse-theorem sections

Result. The eight sections the second pass left unread are now read in full, about 3,600 lines between 25602 and 29449. With the second pass this covers all 17 sections of the inverse-theorem part (23826–30064). In each of the eight I re-derived the argument from the manuscript's own statements and found no error. One step in the frequency analysis omits a normalization it needs. The normalization is available, the manuscript itself uses it 650 lines later, and the conclusion stands. Apart from that, what the internal proof rests on is the list of published statements at the end of this section, which the manuscript imports and I did not re-prove.

Astra's four topics map onto the sections as follows. Frequency analysis is "Sections 9–11 at every rank step" together with Theorem F. Factorization is Theorem D and the equal-scale section. Rank iteration is the degree-conversion section, fed by the petal quotient and the Section 12 construction. Symmetry is the multilinearization section and the Section 13 join.

Every row below has the status **verified** relative to the inputs named in it. Verified here means I reproduced the argument by hand from the text. Nothing in this part can be machine-checked.

Section (lines)	What it supplies	What I re-derived
One-variable factorization and linearization (25602–26076)	Theorem D: $g = \varepsilon \cdot p \cdot \gamma$ with p totally equidistributed in every finite cover of $P/(P \cap \Gamma)$, ε slowly varying, γ periodic modulo Γ. Theorem F: if $X_1(h_1)+X_2(h_2)+X_3(h_3)+X_4(h_4)$ is $O(N^{-j})$ on $\gg N^3$ additive quadruples, then $X_4(h) = \delta + \Sigma\{\alpha_k h\}\beta_k + O(N^{-j})$ on a dense set.	The counterexample to the literal reading of GTZ D.2 (n/N on a half interval; first Fourier average tends to $2i/\pi$). The lattice closed under the Baker–Campbell–Hausdorff product. The period argument. The minimal-dimension descent. All of the finite proof of Theorem F: rounding grid, energy bound, separation by random characters, $4\Gamma'' - 4\Gamma''$ a graph, Bogolyubov, proper progression, fixed carries, exact Freiman property. Relative to GT12 Proposition 9.2 in one variable and GT08 Theorems 5.2, 5.3, Bogolyubov, Lemma 10.4.

Section (lines)	What it supplies	What I re-derived
Equal-scale factorization, bias and torsion (26077–26576)	The factorization on $[N]^k$ in several variables. Replacements for GTZ E.11 and E.13 on the symmetric box $[-N,N]^k$.	The large-offset lemma. Reduction of a dense rectangle to a cube of side ηN. The identity showing p lies in the kernel of the horizontal character. The common quotient P/Δ with $\Delta = P \cap \Gamma_0$, Γ_0 the normal core. The non-zero coset integral. Rationality of a connected subgroup with compact quotient. The exact reconstruction of χ on the whole box from the orbit $(\tilde{p}(n), n/10N, n/q)$. The step $\eta(K) \subseteq \mathbb{Z}$ and K connected, so $\eta(K) = 0$. Relative to the equal-side theorem in the Green–Tao erratum and GT12 Lemma 6.7.

Section (lines)	What it supplies	What I re-derived
Sections 9–11 at every rank step (26577–27137)	The frequency analysis: universal representation; regularization into core frequencies, linear petals {αh}β and independent petals; removal of small and rational frequencies; vanishing of the top character on words with two petals or one independent petal.	The two selection rules (overspill). The three exhaustive cases of the lexicographic descent, each lowering the count. The Freiman property from boxes of width below 1/4. The unit padding in rational removal. The inclusion $\pi_i(P \cap G_i) \supseteq \Xi_i^{\perp}$ with denominators kept. The duality argument for the three spaces V_{123}, V_{124} and V_B. The commutator lifts giving $(\varphi_1(w),1,1,1,1) \in P$. The real-zero conclusion from the one-parameter subgroup. One normalization is missing; see below.
Petal quotient and comparison (27138–27672)	ker q = G* = exp I. Descent of the top character to the quotient, integral on its lattice. The comparison: $\chi_h \otimes \text{conj}\, \tilde{\chi}_h$ has degree–rank strictly below (m,R).	Ideal and filtration properties from additivity of weight, length and multidegree. Triangularity of the bad Hall commutators. The exact lattice image. The comparison filtration as a sum of two subalgebras, and its bracket law. Polynomiality of the triple orbit. Cancellation of the vertical multiplier on the diagonal top group.

Section (lines)	What it supplies	What I re-derived
Section 12 semidirect product (27673–28169)	One nilcharacter $\chi(h,n)$ of multidegree (1,m) that restricts to $\tilde{\chi}_h$ for a dense set of h.	Extension of the character to the full weight-m group. Both group laws, with associativity and inverses. The Lie bracket with its signs. The bigrading and the resulting filtration; nilpotency class at most m+1. The lattice and its cocompactness. The orbit identity that turns αh into its fractional part by right multiplication by a lattice element. The unit completion.
Degree conversion and rank iteration (28170–28435)	Refinement of a degree filtration to a degree–rank filtration. The one-step interface. The m steps from rank m down to rank 0.	$H_(d,0)$ = $H_(d,1)$ = G_d and support at (m,m). The recovery identity, which never divides by a component. Termination: support at (m,0) forces degree at most m−1.

Section (lines)	What it supplies	What I re-derived
Scaling and symmetric multilinearization (28436–28874)	GTZ E.8(v): $[\chi(qn)] = q^{D}\cdot[\chi(n)]$ with D the total degree. Divisibility of symbols. Multilinearization with the factorial retained. Literal symmetry in the last s−1 variables.	Both Rees-algebra maps are Lie maps; for the second this is the identity $a!\cdot b!\cdot\Pi C(a_i+b_i, a_i) = (a+b)!$. The top groups, and the phase cancellation, which requires the tensor power $F^{\otimes C}$ with $C = d!$. The symmetrized product is non-zero and equals the tensor power on the diagonal. The output $[\chi] = s\cdot[\tilde{\chi}(h,n,\ldots,n)]$.
Section 13 symmetry (29025–29449)	Each component of $\chi(h,n)$ is a bounded combination of components of $\Theta(n+h) \otimes \text{conj}\ \Theta(n) \otimes \Psi(h,n)$, with Ψ of lower order.	The three Cauchy–Schwarz steps with the substitution $c = a+b+b'$ in $[-6N,6N]$. The symbol identities (13)–(18), including the coefficient $(-1)^{(s-1)}\cdot s!$. The passage to a convex polytope and one orthant. The pullback giving $s!\cdot(T(h;n^{r}) - T(n;h,n^{r-1})) = 0$. The count $T(h;n^{r}) + r\cdot T(n;h,n^{r-1}) = s\cdot T(h;n^{r})$.

The step that needs one more sentence (lines 26880–26890). After a rational frequency is removed, the representation lives on a finite cover G/Γ'. The text then sends each letter to the M-th power of its lattice element and divides its frequency by M, choosing $\{\alpha h\}\cdot(\beta/M)$ for linear petals. For this to be a representation, the i-th horizontal Taylor coefficient of the orbit must equal the combination of the new letter images with the divided frequencies modulo the smaller lattice $\text{Horiz}_i(\Gamma')$. The earlier steps give the corresponding identity only modulo $\text{Horiz}_i(\Gamma)$. The difference is a lattice vector λ in $\text{Horiz}_i(\Gamma)$, and the divided frequencies work only when λ lies in the span of the letter images plus $\text{Horiz}_i(\Gamma')$. The letter images generate only a sublattice of full rank (line 26729),

so this can fail. A small example: in the Heisenberg group with Γ the integer points, removing the frequency ½ on the letter (1,0,0) gives $\Gamma' = \{(a,b,c): b \text{ even}\}$ and $M = 2$. For a linear petal on the letter (0,1,0), the frequency needed on the cover is $\{\alpha h\}\beta/2 + \lambda_h/2$ with λ_h an integer that the earlier steps do not control, so $\{\alpha h\}\cdot(\beta/2)$ is right only when λ_h is even.

The repair is one sentence. Before removing anything, multiply the orbit on the right by a Γ-valued polynomial sequence, a product of factors γ_i raised to the binomial coefficient $C(n,i)$ with γ_i in $\Gamma \cap G_{(i,1)}$, chosen so that the horizontal identities hold exactly in the vector groups for the chosen real lifts. This does not change the orbit modulo Γ, so the nilcharacter is unchanged, and it moves the i-th horizontal coefficient by any prescribed element of $\mathrm{Horiz}_i(\Gamma)$. Exact equality then survives the removal of small and rational coordinates and the division by M. The manuscript makes exactly this normalization at lines 27527–27534, inside the comparison proposition. I count this as a missing sentence and not as an error.

Interfaces between the sections. I checked that each section's hypotheses are the stated outputs of the section that feeds it: the two analytic contracts in the frequency analysis are Theorems D and F as proved; the two vanishing statements are the hypotheses of the quotient theorem; the one-step premise of the rank iteration is the output of the Section 12 construction and the comparison proposition; the endpoint of the rank iteration is the hypothesis of the symmetry theorem; equation (16) of the multilinearization is equation (3) of the symmetry section; and the bias and torsion theorem is applied in the symmetry section with the dimension, degree and box it was proved for.

Published statements these sections import.

Checked against the source in this pass, and matching the use made of them:

- Green–Tao, *The quantitative behaviour of polynomial orbits on nilmanifolds* (arXiv:0709.3562v6): Definition 1.1 (it requires $G = G_0 = G_1$, which is why the weak factorization step assumes it), Definitions 2.5 and 2.6, Theorem 2.9, Lemma 6.7 (stated for several variables), Definition 8.2, Proposition 9.2, Proposition A.10, Lemmas A.2, A.3, A.11 and A.12. Lemma 8.4 requires offsets of bounded height, so the manuscript's own large-offset lemma is a real supplement and not a restatement.
- Its erratum (arXiv:1311.6170): the one-variable theorems are unaffected, and the several-variable theorem holds when all sides are equal. This is the form the manuscript uses.

- Green–Tao, *An inverse theorem for the Gowers $U^3(G)$ norm* (arXiv:math/0503014): Theorem 5.2 (Balog–Szemerédi–Gowers), Theorem 5.3 (Plünnecke), Lemma 9.2, Lemma 10.4 (Bohr sets contain large proper coset progressions, with the linear-independence clause), and the Bogolyubov lemma. The Bogolyubov lemma is Lemma 6.3 in the arXiv numbering. The manuscript cites it as Lemma 6.4, pp. 97–98 of the journal version, which I could not see, so that label is unconfirmed.

Used as the manuscript states them, not checked against the source: from Green–Tao–Ziegler, Appendix A (ultralimit selection and uniformity), Corollaries B.4 and B.5, Lemmas B.7 and B.9, the definitions of Section 6, Definition 9.6, Lemma 9.12, and E.3, E.5, E.6, E.7, E.8(i)–(iii),(vi); Corollary A.2 of Green–Tao, *Linear equations in primes*; and the published U^3 inverse theorem that starts the induction.

What this does and does not establish. Every section of the internal proof has now been read and its argument reproduced, and the chain from the inverse theorem at one order to the next closes through the stated interfaces. This is one auditor's reading. The steps where an error would be hardest for me to catch are the uniformity and internal-choice steps in the ultraproduct setting; I followed each of them, but no computation can confirm them. The second pass suggested citing the published theorem instead of relying on the internal proof, partly because half of the proof was unread. That reason is gone. The other reason remains: a 6,200-line proof with one reader.

The height-estimate dependency chain

Result. The manuscript has four height estimates, and I traced each from its statement down to the published theorems it uses. I read the current text of all four in full, re-derived every inequality and constant in them, and recomputed the numbers with my own code (`height_checks.py`, 42 checks, all pass). I found no error. Each estimate is conditional on published theorems that I checked in statement and did not re-prove. The deepest of these is the modularity theorem at the prime five (Pan, 2022), which all four use.

Estimate	Statement	Constant	Where it is used
Six-contact (line 707)	$h(u) \leqslant H_6 \cdot M(u)^{24}$ for rational u outside $\{0, -15, -20, -24, -30, -60\}$	Explicit but astronomically large	Only the older existence theorem (line 219)

Estimate	Statement	Constant	Where it is used
Five-contact, exponent 24 (line 7651)	$h(a) \leqslant H \cdot M_5(a)^{24}$ for a outside $\{0, 1, 2401, -343/32\}$	$\log_2 H < 4{,}783{,}227$ claimed; exact bit length 4,645,522	The degree-8 predecessor; its construction is reused by the next two
Five-contact, exponent 19 (line 10005)	$h(a) \leqslant H_{19} \cdot M_5(a)^{19}$	$\log_2 H_{19} < 2{,}093{,}966$ claimed; exact bit length 2,093,448	The 314-variable theorem and the amplification corollary
Effective contact height (line 30714)	$h(a) \leqslant C_3 \cdot M_5(a)^3 \cdot (1 + \log M_5(a))$	$C_3 < 2\text{^}330770$ claimed; my bound $2\text{^}330683$	The 222-variable family

Here h is the logarithmic height, and $M(u)$, $M_5(a)$ are the products of the primes outside a fixed finite set at which the argument meets one of the listed values to odd order.

The five-contact chain, link by link

The exponent-24, exponent-19 and exponent-3 estimates share one construction. Its links, with what I did for each:

1. **The five branch points.** The Shimura curve of discriminant 210 has genus 5, and its quotient by the Atkin–Lehner group is a projective line with five branch points of index two, normalized to ∞, 0, 1, 2401, −343/32. The input is the table of four genus-one quotients in González–Rotger. **Verified.** The four quartics in the manuscript are the four rows for discriminant 210 in the source. My code recomputes their j-invariants, confirms the Legendre parameters 1/2401, 32/375, 1/225 and 5/1029, and finds that (2401, −343/32) is the only pair consistent with all four. The Riemann–Hurwitz count that gives exactly five branch points is correct.
2. **The level-three model and the field of a fibre.** A fine moduli scheme with full level-three structure over $\mathbb{Z}[1/210]$, the group of order 576 acting on its fibres, and the conclusion that a fibre over a rational point is defined over a normal field L of degree dividing 576, unramified outside the primes of the contact product, with good reduction there. **Followed, not re-derived.** I re-derived the counts (72 units modulo 3, degree 36 over the coarse curve, 36·16 = 576) and the torsion-freeness argument. The identification with the canonical model rests on Milne's notes on Shimura varieties and on Deligne, which I did not check.

3. **The geometric comparison** $h(a) \leqslant 2 \cdot \max(0, h_F(A)) + 128$, where h_F is the stable Faltings height of the fibre surface. The input is Yuan's theorem that the square of the Hodge bundle is 210 times the square of the canonical bundle, isometrically. **Arithmetic verified; the curvature comparison followed.** The source statement matches, including the metrics and the regular semistable model. The shift by $\log \pi$ between the two height normalizations is right. The constants $J = 2{,}765{,}952{,}000$ and $C_{met} = 12{,}540{,}153{,}541{,}944{,}538{,}512$ and the bound 2^{160} on the additive term are right.
4. **Fibres with complex multiplication:** $h_F(A) < 854 \cdot R$. **Verified**, given the isogeny theorem of Gaudron–Rémond (2023).
5. **Fibres without complex multiplication.** In order: potentially good reduction (re-derived); the descent of the surface to a two-dimensional Galois representation of the rationals, with a sign cocycle split by a Dirichlet character of controlled order, giving a field L' of degree at most $2304 \cdot R$ (re-derived); modularity of that representation (the lemma at line 506: followed, with each hypothesis of the modularity theorem checked on the actual representation); the level equals the conductor (Carayol); the conductor bound $N \leqslant 210^{300} \cdot R^2$ (re-derived: last upper break at most 136, exponent at most 274); and the transfer of the height from a modular Jacobian by an isogeny estimate (re-derived). **Verified as a derivation from the cited theorems.**
6. **The three endings.** Exponent 24 uses the Jacobian of $X_1(N)$. Exponent 19 uses the quotient by the kernel of the nebentypus, whose degree over the j-line is at most $9216 \cdot 210^{300} \cdot R^{(19/6)}$; the sixth power gives R^{19}. Exponent 3 replaces the general height bound for Belyi curves by the manuscript's own bound for these modular curves (verified in the first pass on Version 15). **Verified.**

The passage from the exponent-19 estimate to the heights used in the 314-variable argument (line 10219: $h(a_i) < 2 \cdot H \cdot |N|^{(19/e)} + 14$, then $h(x(Q)) < 7 \cdot H \cdot |N|^{(19/e)}$) is correct.

The six-contact chain

This estimate has the same shape with the Shimura curve of discriminant 6. I read lines 473–1705 in full, which the earlier passes had not done for the Version 15 text. **Arithmetic verified; the moduli-theoretic construction followed.** In particular the identity $G^6 - 97200 \cdot F^2 = H^2$ behind the pullback, the Igusa-invariant normalization (which agrees with the formula published by Baba and Granath), the degree bound 324, the constant $3{,}779{,}136 = 2^6 \cdot 3^{10}$, both branches of the height bound, and the final constant 10^{13} are right.

Published inputs of the chain

Checked against the source, in this pass or an earlier one, and matching the use made of them:

- González–Rotger, *Non-elliptic Shimura curves of genus one*, Table 2 (this pass).
- Yuan, *Explicit Kodaira–Spencer map over Shimura curves*, Theorem 1.1 and Section 2.1, in the English version on arXiv (this pass). The manuscript cites the published Chinese article, which I did not see.
- Pazuki, *Theta height and Faltings height*, Corollary 1.3 and its normalizations (this pass).
- Baba–Granath, the Igusa invariants on the discriminant-6 locus (this pass).
- Gaudron–Rémond 2014, Theorem 1.4, and 2023, Theorem 1.9(1) and Lemma 16.2; Javanpeykar, Theorem 1.1.1 and Propositions 4.1.1 and 4.1.3; von Känel, (2.2), (4.1), (4.2), (5.9); Carayol 1987; Fowler, Proposition 2.7; Rémond, Corollary 1.3; Pan, Theorem 1.0.4 (earlier passes).

Used as the manuscript states them, not checked against the source: Calegari's survey, p. 624, through which the modularity theorem is cited; Faltings; Scholze; Deligne–Serre; Kato (2004), §9.4 and (14.10); Shimura; Li; Buzzard; Milne; Deligne (1971, 1979); Lin–Yang; Igusa (1967); Silverberg–Zarhin; Mascot; Diamond–Im; and the height formula for elliptic curves with complex multiplication.

What the chain is conditional on

- **Modularity at the prime five.** Every non-CM fibre is handled through Pan's theorem. It is published (Journal of the AMS, 2022). I checked that the manuscript verifies its hypotheses for the representation it applies it to. I did not audit the theorem.
- **Yuan's isometry.** It carries the whole comparison between the height of a point and the Faltings height of its fibre. The statement matches. I did not audit its proof.
- **The canonical-model identification** of the level-three curve, which gives the degree 576 and the ramification control. I followed it and did not re-derive it.
- The isogeny, Belyi-height and conductor theorems are standard published results used inside their stated hypotheses.

The citation slip reported under "Sources examined directly" (two results of Gaudron–Rémond attributed to the wrong paper, lines 30633 and 30646) is in this chain and does not affect it.

The simultaneous-primes dependency chain

Result. The theorem at line 17565 produces integers u, w for which all of $u + c_j \cdot w^2$ are prime, with Gaussian generators in prescribed residue classes and with the neighbouring norms $N(\pi_j - 1)$, $N(\pi_j + 1)$ sharing no odd prime. Its proof is the section 17544–20351, about 2,800 lines. Earlier passes had checked only its two ends. I have now read all of it. I re-derived the algebraic identities, the sieve estimates, the exponent bookkeeping, the ray-class computation and the final order of limits. I followed the four steps that run through published transference machinery. I checked the statements of the main published inputs against their sources. I found no error. The theorem is conditional on those inputs and on the internal inverse theorem at every order the transfer calls for.

Step (lines)	What it proves	Published input	Status
Local counts (17597–17648)	Point counts on the quadric $a^2+b^2-c^2-d^2 = \kappa w^2$ over finite fields; local sieve density $(k + 2\chi(\ell))/\ell$	None	Verified
Primitive points with congruence conditions (17649–17694)	A main term of size B^3 times the product over ℓ of $\#\Omega(\ell)/(\ell^4-1)$, with error B^(5/2+ε)·m^(5/2+ε)	Browning–Loughran, Theorem 4.1	Verified, given the input. The passage from their count to primitive points is right.
Medium primes (17695–17773)	An upper-bound sieve of dimension k for common divisors between L and B^(1/100)	Williams (Mertens in progressions)	Verified, given the input
Large primes (17774–17923)	Common divisors above B^(1/100) on the quadric	Browning–Heath-Brown, Lemmas 2.1 and 4.1	Verified, given the inputs. I re-derived the change of variables and both polynomial identities by hand.

Step (lines)	What it proves	Published input	Status
Obstruction tail (17924–17985)	The weighted count of patterns with a forbidden common divisor above L is at most CτN^(3/2)/(L log L) plus a power saving	None	Verified
Ray-class weights (17986–18276)	Main terms of the ray-class prime weight on progressions; the centered weight has negligible sums on progressions	Mitsui 1956 (prime ideal theorem with characters of small conductor)	Verified, given the input
Equidistribution along Gaussian norms (18277–18510)	A totally equidistributed orbit stays equidistributed along the values of x^2+y^2 in residue classes	Green–Tao, Theorem 2.9 and Lemma 6.7	Verified
Type I and II sums on ideals (18511–18990)	A Vaughan identity for Gaussian ideals; minor-arc bound with a fixed power of the equidistribution parameter	Green–Tao, Lemma 4.5	Verified
Affine interpolation (18991–19205)	The same bound for the weight restricted to Wn + b	Green–Tao, Theorem 2.9	Verified
Factorization (19206–19441)	The centered ray-prime weight is orthogonal to every nilsequence of bounded complexity, with any power of log N saved, for each fixed modulus	Green–Tao, Theorem 1.19; a conjugation estimate from their Möbius paper	Verified, given the inputs

Step (lines)	What it proves	Published input	Status
Majorant and inverse step (19442–19742)	The centered weight has small Gowers norms of every fixed order	Green–Tao, *Linear equations in primes*, Propositions 6.4 and 10.1; the internal inverse theorem	Followed. The use of Proposition 10.1 matches its statement exactly. *Fourth pass: the majorant lemma is now compared with the published Appendix D and verified, given its estimates.*
Linear replacement (19743–19835)	In any fixed system of linear forms, ray-class prime weights can be replaced by ordinary prime weights times their densities	Green–Tao, Proposition 7.1 and Lemma 4.4	Followed
Local cube (19836–19887)	The averaged global Gowers norm needed by the polynomial method is small	Tao–Ziegler, Proposition 6; Bienvenu	Followed. The expansion of the centered cube is right.
Polynomial transfer (19888–20165)	The asymptotic count of the patterns $u + c_j \cdot w^2$ with Frobenius conditions	Tao–Ziegler, Theorems 9, 11, 13, 15 and Corollary 3	Followed. Parameters match the source.
Ray-class lower bound (20166–20304)	The singular series is at least c·(log L)^(−4k)	Kopp–Lagarias	Verified
Order of limits (20305–20351)	The theorem	None	Verified

Points to note

- **Two published results are used through their proofs.** Tao and Ziegler state their theorems for the modulus W equal to the product of the primes up to a cutoff. The manuscript uses W with a fixed extra factor, the conductor, and says the divisor-sum construction is unchanged (lines 20063–20073). The same happens with Green–Tao's Proposition 6.4, where the manuscript writes the changes out as its own lemma (19498–19642). Both adaptations look right to me. I did not check them line by line against the published proofs, which is why these steps are marked followed.
- **The inverse theorem enters here.** The step from nilsequences to Gowers norms uses the internal inverse theorem at order max(s, 4) − 1 for each norm order s that the transfer needs (lines 19474–19476 and 19704). So this theorem, and with it the 314-variable theorem and the 222-variable family, inherit the status reported under "The remaining inverse-theorem sections".
- **Nothing here is effective**, and the manuscript does not claim it is. The constant in Mitsui's estimate and the inverse theorem give no bounds. The theorem is used only for existence.
- **A second prime input is outside this chain.** The Green–Tao theorem on linear forms in primes is applied at line 2217, with the same inverse theorem. I did not re-audit that application in this pass. *It is verified in the fourth pass.*

Published inputs

Checked against the source in this pass, and matching the use made of them:

- Green–Tao, *Linear equations in primes* (arXiv:math/0606088): the definitions of the linear-forms and correlation conditions, Proposition 6.4, Proposition 7.1 and Proposition 10.1. Proposition 10.1 assumes the inverse conjecture at order s and a measure that is (s+2)·2^(s+1)-pseudorandom, and returns correlation with a linear nilsequence from a finite family, which is how the manuscript uses it.
- Tao–Ziegler, *Polynomial patterns in the primes* (arXiv:1603.07817v1): Theorems 1.4, 2.5, 3.1 and 4.1, Corollary 3.9, Proposition 3.10 and Theorem A.1, which are Theorems 4, 9, 11 and 13, Corollary 3, Proposition 6 and Theorem 15 of the journal version. The dense-model theorem is stated for an arbitrary function bounded by the prime weight, as the manuscript says.
- Browning–Loughran, *Sieving rational points on varieties* (arXiv:1705.01999), Theorem 4.1.

- Browning–Heath-Brown, *The geometric sieve for quadrics* (arXiv:2003.09593), Lemmas 2.1 and 4.1.
- Green–Tao, *The quantitative behaviour of polynomial orbits on nilmanifolds*: Definition 1.2, Theorem 1.19, Theorem 2.9, Lemma 4.5 and Lemma 6.7.

Not checked against the source: Mitsui (1956), Lemmas 14 and 15; Kopp–Lagarias; Williams; Bienvenu, Theorem 1.1; the conjugation estimate on p. 547 of Green–Tao's Möbius paper; Lemma 4.4 and Appendices C and D of *Linear equations in primes*. *All except Mitsui were compared in the fourth pass.*

Findings for the third pass

No error found. Every inference I examined in this pass holds, and every constant and identity I recomputed agrees with the text.

What changed since the second pass

1. **The baseline charts are proved**, where before they were only tested at sample points.
2. **The internal inverse theorem has been read in full.** All 17 sections are now verified or followed. The second pass had covered about 2,700 of 6,240 lines.
3. **The height estimates are re-derived**, including the constant H_{19} behind the 314-variable theorem and the constant C_3 behind the 222-variable family. The exact sizes are below the stated bounds.
4. **The simultaneous-primes theorem is traced end to end.** Twelve of its sixteen steps are verified and four are followed.

Points for the authors

None of these affects a result.

- **Inverse theorem, lines 26880–26890.** The passage to a finite cover divides each frequency by M without first making the horizontal identities exact. One sentence repairs it, and the manuscript already uses the same normalization at lines 27527–27534. Details are under "The remaining inverse-theorem sections".
- **Lines 30633 and 30646.** Two results of Gaudron and Rémond are cited from their Duke paper. They are in their paper in *Commentarii Mathematici Helvetici* 89 (2014), which is not in the bibliography.

- **Line 25988.** The Bogolyubov lemma is cited as Lemma 6.4 of the Green–Tao U^3 paper. It is Lemma 6.3 in the arXiv version. I could not see the journal numbering, so this needs a look and may be correct. *Resolved in the fourth pass: the journal numbers it 6.4, on pp. 97–98. The citation is right and no change is needed.*
- **Lines 20063–20073.** Tao and Ziegler's theorems are applied with a modulus that has a fixed extra factor. The text says their proofs are unchanged. A sentence saying this is an adaptation of the proofs, and not an application of the statements, would make the dependence clear.

What the results now rest on

For the two headline results, the 314-variable theorem and the 222-variable family, the parts of the manuscript examined across the three passes hold. What they are conditional on is outside the manuscript, plus the items this audit has not reached:

1. **Published theorems used as stated.** The heaviest are modularity at the prime five (Pan), Yuan's isometry for the Hodge bundle on Shimura curves, the isogeny theorems of Gaudron and Rémond, the Green–Tao transference results, the Tao–Ziegler polynomial method, and Mitsui's prime ideal theorem. I checked their statements, except Mitsui's, and did not audit their proofs.
2. **The internal inverse theorem**, which the prime-selection theorems need. It has had one complete reading, mine. The statement is also in the literature.
3. **Kato 1999 and Ribet 1990**, which I have not read. The first carries the repaired application at $p = 2$ reported in the second pass.
4. **Inputs no pass has re-audited**: Kolyvagin's theorem with modularity, the formal-group and Mordell–Weil facts, and the application of the Green–Tao theorem at line 2217. *All four are covered in the fourth pass.*

What I would do next

- Attach PDFs of Kato 1999 (pp. 314–318) and Ribet 1990 (pp. 440–443), and I will finish those two checks.
- Decide whether the inverse theorem is cited or proved internally. The internal proof now has a complete reading behind it; a second reader who works in this area would still be worth having for a proof of this length.
- If the remaining inputs in item 4 matter to you, they are the next candidates for a pass.

Fourth pass: finishing the audit (11 Nov)

Scope and result

Result. This pass covers what the third pass left open: the Mordell–Weil generator, the Kolyvagin and modularity inputs, the formal-group inputs, the Green–Tao application at line 2217, the sources not yet compared, the comparisons that had used arXiv versions, and Astra's four follow-ups. I found no error. One point I raised in the third pass is withdrawn: the Bogolyubov lemma is cited correctly. Kato 1999 and Ribet 1990 are still unread, because I could not open either from this session.

Item	What I did	Outcome
Mordell–Weil generator (lines 363–446)	Re-derived the descent, the torsion, the height comparison and the enumeration, by hand and with my own code	**Verified.** $E(K) = \mathbb{Z}R_0 + E[2]$ and $2E(K) = \mathbb{Z}P$
Kolyvagin and modularity (lines 1832–1880)	Re-derived the lemma and compared Kolyvagin's Theorem 1 and Kato's Corollary 14.3 with the sources	**Verified** as a derivation from the cited theorems. The statements match
Formal-group inputs (all uses)	Re-derived each use and recomputed every explicit value	**Verified**
Green–Tao application (line 2217, and the parallel one at 9561)	Compared the hypotheses with the published theorem and re-derived the rest of the proposition at 2071–2530	**Verified**, given the Green–Tao theorem and its two inputs
Other sources	Compared 22 sources with the text, in the published version where I could reach it	All match. Those not reached are listed

Item	What I did	Outcome
Astra's four follow-ups	See the next section	1: repair re-checked, text supplied. 2: exact entry supplied. 3: no change needed. 4: adaptation confirmed in outline, alternative supplied
Kato 1999 and Ribet 1990	Could not be opened	**Not read**

Terms. As before. *Verified* means I re-derived the step by hand or reproduced it with my own code. *Followed* means I read the argument and found no gap, without re-deriving it. *Not audited* means I did not check it. A result is *conditional* when it rests on something this audit did not validate; that describes the audit, not the proof.

Method. I ran no code from the repository; the sandbox does not allow it. The computations are in two scripts of mine with exact arithmetic: `mw_checks.py` (23 checks) for the Mordell–Weil generator and `fg_checks.py` (48 checks) for the formal-group, anchor and Lucas inputs. All pass. On its first run one check in the second script failed. The fault was in my test, a witness search that did not allow the denominator 5, and not in the manuscript. I corrected the test and added three local checks.

Astra's four follow-ups

1. The horizontal normalization before the finite cover

Result. I re-derived the repair in the manuscript's notation and it holds. It takes one inserted paragraph and one added sentence.

Why it is needed. At line 26731 the frequencies ξ_a are chosen so that the horizontal Taylor coefficient of g_h equals $\Sigma_a \xi_a \cdot \varphi(e_a)$ modulo the horizontal lattice of Γ. At lines 26880–26890 each letter is sent to its M-th power and each frequency is divided by M. Dividing a frequency means dividing a real lift x_a of it. Since $\Sigma_a (x_a/M)\cdot(M\cdot\varphi(e_a)) = \Sigma_a x_a\cdot\varphi(e_a)$, the new identity holds modulo the horizontal lattice of Γ' only if the old identity held for these lifts as an equality of vectors. As written it is known only modulo the horizontal lattice of Γ, which is larger. The Heisenberg example in the third pass shows that the difference can be nonzero.

Insertion. At line 26815, after the first sentence of the subsection “Removing rational and small coordinates without losing unit norm”:

> Before removing anything, make the horizontal identities exact. Fix internal real lifts x_{h,i,a} of the regularized frequencies: the small lift for a small coordinate, a rational lift for a rational coordinate, and {αh}β for a linear petal. For each i, the difference between the i-th horizontal Taylor coefficient of g_h and Σ_a x_{h,i,a}·φ(e_{i,a}) lies in the horizontal lattice of Γ, so it is the image of some γ_{h,i} in Γ ∩ G_{(i,1)}. Replace g_h(n) by g_h(n)·Π_i γ_{h,i}^(−C(n,i)), where C(n,i) is the binomial coefficient. This does not change the orbit modulo Γ. The factor is polynomial by the Taylor criterion B.9, and the horizontal Taylor map is a homomorphism, so the horizontal identities now hold in the vector groups themselves. In the two removals below, use these lifts as the exponents. Each removal then subtracts the removed term exactly, and the identities persist for the remaining coordinates.

Added sentence. At line 26883, after “divide its horizontal frequency by M”:

> Divide the fixed lift. Since Σ_a $(x_a/M)\cdot M\cdot\varphi(e_a) = \Sigma_a$ $x_a\cdot\varphi(e_a)$, the vector identities are unchanged, and so they hold modulo the horizontal lattice of Γ′.

What I checked.

- The element γ_{h,i} exists, because the horizontal lattice is by definition the image of Γ ∩ G_{(i,1)}.
- The orbit is unchanged, because the factor takes values in Γ.
- Polynomiality and the homomorphism property are the manuscript’s own statements at lines 26735–26742.
- The manuscript’s three ways of dividing (the value {αh}(β/M) for linear petals, fixed roots for core coordinates, internal roots for independent coordinates) are all divisions of a lift, so they agree with the added sentence.
- The integer change of basis at lines 26746–26755 comes first, so lifts of the new frequencies determine lifts of the old ones.
- A rational coordinate has a rational lift and a small coordinate has its small lift, so both removal steps subtract exactly the term that appears in the identity.
- The lifts and the γ_{h,i} are internal choices, as in the manuscript’s own use of this device at lines 27527–27534.

2. The two Gaudron–Rémond citations

Lines 30633 and 30646 should cite a new bibliography entry:

> É. Gaudron and G. Rémond, *Théorème des périodes et degrés minimaux d'isogénies*, Commentarii Mathematici Helvetici 89 (2014), no. 2, 343–403, doi:10.4171/CMH/322. §2.3 and Corollaire 8.4.

The existing entry for the Duke paper stays for Theorem 1.4 and Section 9 (lines 1322, 1599 and 1616). I confirmed the volume, pages and DOI on the publisher's page.

3. The Bogolyubov lemma numbering

Resolved. No change is needed. In the published paper (Proc. Edinburgh Math. Soc. 51 (2008), 73–153), Section 6 is numbered: Proposition 6.1, Lemma 6.2, Remark 6.3, Lemma 6.4 (the Bogolyubov lemma, pp. 97–98), Remark 6.5, Proposition 6.6. The arXiv version leaves the remarks unnumbered, which is why the lemma is 6.3 there. The manuscript's "Lemma 6.4, pp. 97–98" at line 25988 is correct, and I withdraw the point.

The same published text confirms the other labels the manuscript takes from this paper: Proposition 3.2 (p. 88), Theorems 5.2 and 5.3 (p. 95), Lemma 9.2 (p. 113), Lemma 10.4 with its linear-independence clause (p. 121), and Theorem 10.11 (p. 124). It also confirms the misprint the manuscript reports in Example 12.3 (p. 136): the printed iterate has ½·mn(n+1)·α and no term mnx, where direct iteration gives mnx + mα·C(n,2).

4. The Tao–Ziegler application

Result. Astra's description is accurate: lines 20063–20073 adapt the proofs and do not apply the statements. Each change the manuscript claims is correct as far as I can check it against the published paper. There is also a way to avoid the adaptation.

The published statements. Equations (32)–(34) set W equal to the product of the primes up to w, A = W^(1/κ) and R = N^κ, and define the weight $x \mapsto (\varphi(W)/W)\cdot\Lambda'(Wx + b)$ for a residue b coprime to W. Call it the weight for (b, W). Theorem 11 and Corollary 3 hold for an arbitrary function f between 0 and the weight for (b, W). Theorem 9 concerns bounded functions and does not involve W. Theorem 13 allows functions bounded by 1 plus the weight for (b_i, W), with different residues b_i. Theorem 15 is the polynomial forms condition for the majorant. The numbering is the one the manuscript cites. The manuscript's modulus is D times the primes up to v that do not divide D, so the theorems that involve W do not apply to it word for word.

The three claims at lines 20063–20073.

1. A unit residue makes every divisibility condition at a prime dividing W false, also at the extra prime powers. True: Wx + b is congruent to b modulo every such prime.
2. The normalization is unchanged. True: φ(W)/W depends only on the set of primes dividing W, and once v exceeds the primes of D that set is the set of primes up to v.
3. The local factors at the other primes are unchanged and the coefficient bounds absorb the fixed factor. True at the level I could check: W is a unit modulo those primes, and in the proof of Theorem 15 the exceptional primes are the primes above w that divide a difference $(WP_i + b_i) - (WP_j + b_j)$, whose number is controlled by the coefficient bound.

I did not re-read, with the changed modulus, the proofs of Theorem 15, Propositions 2 and 3 and Lemmas 4 and 5, or the earlier Tao–Ziegler paper they rest on. So this step stays *followed*.

A route that uses only the statements. Let W_0 be the product of the primes up to v, with v at least the largest prime of D. Then the manuscript's W equals $E \cdot W_0$ with E = D/rad(D) a fixed integer, and $\varphi(W)/W = \varphi(W_0)/W_0$. Write $b = W_0 \cdot b_1 + b_0$ with $0 \leqslant b_0 < W_0$. Then $Wx + b = W_0 \cdot (Ex + b_1) + b_0$, so the manuscript's weight at x is Tao–Ziegler's own weight for (b_0, W_0) at the point $y = Ex + b_1$. Given f below the manuscript's weight, let $F(y) = f(x)$ when $y = Ex + b_1$, and $F(y) = 0$ on the other residue classes modulo E. Then F lies between 0 and the weight for (b_0, W_0), and Theorems 11 and 13 and Corollary 3 apply to F as published. A polynomial average of functions f_j equals E times the corresponding average of the F_j, taken along the polynomials $E \cdot R_j(m)$ plus a constant, which have integer coefficients and the same degrees. The model to compare F_j with is μ_j times the indicator of its residue class. The global norm of that difference along $q \cdot [J]$ is a fixed multiple of the global norm of $f_j - \mu_j$ along $(q/g) \cdot [J \cdot g/E]$, with $g = \gcd(q, E)$, and the manuscript's estimate at line 19849 covers that range. I checked this outline step by step against the published statements. I have not written it out in full, and it would need that before it replaced the present text.

The Mordell–Weil generator

Result. The proposition at lines 363–446 is verified in full. I re-derived each step by hand and reproduced every number with `mw_checks.py`.

- **The points.** $R_0 = (1 + \sqrt{2}, 2\sqrt{2})$ lies on $y^2 = x^3 - \delta^2 x$ with $\delta = 3 - \sqrt{2}$, and $2R_0 = P$ with the displayed coordinates. The older model has $d = 75 - 53\sqrt{2} = \delta \cdot \varepsilon^{-4}$ with $\varepsilon = 1 + \sqrt{2}$, and the unit change of variables carries P to the older point P_0.

- **Rank.** The image of the descent map for the 2-isogeny is generated by −1, ε and d and has order 8. The image for the dual isogeny is trivial: its classes are totally positive with even valuations, and −4 is not a square modulo 7. So E(K)/2E(K) has order 8, and the rank is 1 once the torsion is known.
- **Torsion.** The reductions at the primes above 17 and 23 with $\sqrt{2} \equiv 6$ and $\sqrt{2} \equiv 5$ have 20 and 24 points. The torsion subgroup injects into both, so its order divides 4, and it is E[2].
- **The generator.** The abscissa class of R_0 has mixed signs at the two real places, so R_0 is an odd multiple of a generator modulo torsion. The duplication bounds $4h(x(Q)) - \log 392 \leqslant h(x(2Q)) \leqslant 4h(x(Q)) + \log 676$ hold: the common factor of the two duplication forms has valuation at most 6 at $(\sqrt{2})$, at most 4 at (δ) and none elsewhere, which I confirmed by brute force over 10,764 primitive pairs. With $h(x(2R_0)) = \log 41$ and the inequality $41^3 \cdot 676 \cdot 392^{36} < 9^{108}$, a point of which R_0 is a proper odd multiple would have its abscissa in an explicit finite list. My enumeration reproduces the manuscript's counts exactly: 7,004 quadratics, 14,008 roots and 111 rationals. The abscissae that give points are 0, $\pm\delta$, $1 + \sqrt{2}$, $23 - 17\sqrt{2}$, $2 + 4\sqrt{2}$ and $(5 - 4\sqrt{2})/2$. These are the points of E[2] and the four translates of R_0 by E[2]. So $E(K) = \mathbb{Z}R_0 + E[2]$ and $2E(K) = \mathbb{Z}P$.
- **The height inequality.** The bound $h(x(nP)) \geqslant n^2 - 6$, which every later section uses, follows from the same duplication bounds.

Kolyvagin and modularity

Result. The lemma at lines 1832–1880 is verified as a derivation from the modularity theorem and Kolyvagin's theorem. The two source statements I could reach say what the manuscript takes from them.

- **The lemma.** For an elliptic curve A over $\mathbb{Q}$ whose L-function vanishes to order $r \leqslant 1$ at $s = 1$, the rank is r and the Tate–Shafarevich group is finite, so the 2-primary Selmer group has corank r. I re-derived the corank step from the Kummer sequence, the remark that the restriction kernel on Tate–Shafarevich groups for a quadratic extension is killed by 2, and the surjectivity of the Picard-norm map from $J_1(N)$ to A, including the supporting sections at lines 30065–30153.
- **Kolyvagin.** His Theorem 1 states that the rank equals the analytic rank and the Tate–Shafarevich group is finite whenever the analytic rank is at most 1. Theorems 2–5 are the steps the manuscript describes. I read this in the Max Planck preprint 1990-69, which is the text of the ICM address under the same title. I did not open the printed proceedings pages.

- **Kato (2004), Corollary 14.3.** For an elliptic curve over $\mathbb{Q}$ with a surjection from $J_1(N)$, an abelian field and a character χ with $L(A, \chi, 1) \neq 0$, the χ-parts of the Mordell–Weil group and of the Tate–Shafarevich group are finite. There is no condition on the prime and no hypothesis on the residual representation. This is the manuscript's independent route for $r = 0$.
- **Around the pole predicate.** The corank lemma at line 1812 is re-derived: with full rational 2-torsion and a 2-Selmer group of dimension 3, the corank is 1. Its inputs match the sources: Poonen–Stoll, Corollaries 4 and 7 and Theorem 8 (for an elliptic curve the pairing on the non-divisible quotient is alternating and nondegenerate), and Dokchitser–Dokchitser, Theorem 1.4 (the p-parity conjecture for every elliptic curve over $\mathbb{Q}$ and every p).
- **What this is conditional on.** The modularity theorem of Breuil, Conrad, Diamond and Taylor, and Kolyvagin's theorem, which itself uses the Gross–Zagier formula and the analytic non-vanishing theorems it cites. I did not audit those proofs. The three statements taken from Silverman's book (C.13.6, II.3.6, III.3.4) are standard; I did not have the book.

Formal-group inputs

Result. Every use of the formal group that I found is verified. I did not have Silverman's book, and nothing below depends on its wording: I re-proved each fact where it is used and recomputed each explicit value with `fg_checks.py`.

- **Conventions (lines 1909–1935).** With $z = -x/y$ and $w = 1/x$ on $y^2 = x^3 - d^2x$, one has $w = 2z^2/(1 + \sqrt{}(1 + 4d^2z^4))$, the invariant differential is $dz/\sqrt{}(1 + 4d^2z^4)$, and the logarithm is the stated series. The logarithm is an isometry where $v(z) > e/(p - 1)$, because the p-adic valuation of r is at most $(r - 1)/(p - 1)$ for every $r \geqslant 1$. Hence $v(z([n]Q)) = v(z(Q)) + v(n)$ there.
- **The logarithm lemmas (lines 10292, 11914 and 31712).** The identity $\log = z \cdot G(D \cdot f^2)$ with $f = 1/x$, the equivalence of the recurrence with the differential equation $4w(1 - w)G' + (1 + w)G = 1 - w$, the valuation $(4m + 1) \cdot r$ of the truncation error, and both bounds on the denominators of the coefficients. I confirmed the series identity to order 60 by exact power-series arithmetic. The global non-vanishing lemma at line 10358 is re-derived from the 2-adic valuation of the coefficients.
- **The anchors (lines 9140–9335 and 12073).** At the prime 3 the point P reduces to the point $(\sqrt{}2, 0)$ of order 2, and $z(2P)$ is the displayed number. For even n, $v_3(z(nP)) = 1 + v_3(n)$ and $v_3(f(nP)) = 2 + 2v_3(n)$; I also checked both exactly for $n \leqslant 40$. Odd multiples

are not in the kernel of reduction. At 2 the point lies on the formal branch, $v(z(nP)) = 3/2 + v_2(n)$, and the coefficients of $f(nP)$ are 2-integral. The seven-witness predicate defines the intersection of the local rings at 2 and 3 (Koenigsmann, Proposition 6, with the Hilbert symbols recomputed). The three-witness anchor is re-derived: the quadratic-embedding criterion (Voight, Proposition 14.6.7), the two local square conditions, which I checked on 1,958 rationals, and the form $A^2 + AB - B^2$. All the numerical cutoffs are right.

- **The local comparison lemmas.** One place for every test (line 12121), the two-anchor comparison (12163), the Version 9 depth argument (20841–20875), the Version 12 mixed-precision lemma (21110) and the 222-family version (32585–32610) are re-derived, including the remark that translation by (0, 0) sends z to −z.
- **Infinite order from the 2-adic branch (lines 31547–31566).** Re-derived.
- **Version 9 arithmetic (lines 20634–20925).** The product-height identity, the odd chart, the separation lemma and the deeper completeness step are re-derived. I also checked the common sign of the valuations at 16 places for odd multiples up to 13.
- **The Lucas growth lemma (line 9832).** Re-derived, and confirmed by brute force for the indices 7, −7, 8, −8 and 9.
- **The Frobenius facts for prime selection (lines 2366–2412).** For $p \equiv 1 \bmod 8$ and $p \equiv 3 \bmod 7$ the two reduced group orders are $p + 1 - 2a$ and $p + 1 + 2a$ with $a^2 + b^2 = p$, the 2-part of their gcd is 4, and p divides neither. Re-derived, and checked at seven primes.

One use is outside this pass: the formal groups of abelian varieties in the restricted-converse section (lines 3972–3992 and 4100–4116). I read that step in an earlier pass and did not re-read it now.

The Green–Tao application at line 2217

Result. Verified. The proposition that contains it (lines 2071–2530, the fixed twenty-two-prime pattern) is now verified in full, given the Green–Tao theorem with its two inputs, Mertens' theorem in progressions and Dirichlet's theorem. Earlier passes had not audited it.

- **The theorem as published.** The Main Theorem (p. 1763) gives the generalized Hardy–Littlewood asymptotic for every system of complexity at most s, assuming the inverse conjecture GI(s) and the Möbius conjecture MN(s). The asymptotic (Conjecture 1.2, p. 1757) is for integral affine forms of bounded size on a convex body inside [−N, N]^d. Complexity is Definition 1.5, and the proof of Lemma 1.6 (p. 1761) notes that a system

with no two forms affinely dependent has complexity at most $t - 2$, by the partition into singletons. MN(s) is Theorem 1.1 of Green–Tao's Möbius paper (p. 542). GI(s) is supplied by the manuscript's internal proof.

- **The hypotheses.** The 22 forms have integer coefficients, each has a coefficient ±1, and any two are independent over $\mathbb{Q}$ and modulo every prime above 324, because the relevant minors are ±1 or contact differences of size at most 61. The region is a box, the coefficients are fixed before X grows, and the complexity is at most 20. So GI(20) and MN(20) suffice, as the manuscript says.
- **The rest of the proof.** I re-derived the local factors, the lower bound $\log \beta \geqslant -484/q^2$ (it needs only that each form is non-constant modulo q), the density of admissible residues, the removal of prime powers, the Selberg weights and both lower bounds for the sieve denominator, the count of prime values with a common odd prime in the two group orders, the count on each fibre, and the order in which L and X are chosen.
- **The second application (line 9561).** The same theorem with at most 41 forms and complexity at most 39. I checked its hypotheses and the local algebra at lines 9525–9612. Its sieve estimates repeat those above.

The majorant lemma in the primes chain (lines 19498–19642). The third pass marked this step followed because I had not compared it with the published proof. I have now compared it with Appendix D of *Linear equations in primes*: Theorem D.3 and its error term, the sieve factor of Lemma D.2, and the proof of Proposition 6.4 (the definition of the measure, the local factors at small and large primes, the exceptional primes, the correlation weight and its moments). Each change the manuscript makes for the fixed modulus factor is correct. In particular the lattice-point error does not depend on the coefficients of the forms, so the sieve exponent can stay fixed, and the domination constant comes out as stated. I did not re-derive the Euler-product estimates (D.4)–(D.23) themselves, which the manuscript uses unchanged.

Sources compared in this pass

Result. Twenty-two sources were compared with what the manuscript takes from them. They agree wherever the text was legible, and two rows say what a scan did not let me read. Where a published version was reachable I used it, and for the papers that the third pass had read on arXiv the published numbering is now confirmed.

Source	Version seen	What was compared	Outcome
Green–Tao, Linear equations in primes	Published (Annals 171)	Main Theorem, Conjecture 1.2, Definition 1.5, Lemmas 1.6 and 4.4, Definitions 6.1–6.3, Propositions 6.4, 7.1, 10.1, 11.2 and 11.5, Corollary A.2, Theorem D.3, Lemma D.2, Proposition D.4	Match, including page 1763 for the Main Theorem
Green–Tao, Möbius and nilsequences	Published (Annals 175)	Theorem 1.1 (p. 542) and the claim on p. 547	Match
Green–Tao, polynomial orbits	Published (Annals 175)	The sixteen numbered items the manuscript uses	Match. Numbering is the same as on arXiv
Green–Tao, U^3 inverse theorem	Published (Proc. Edinburgh Math. Soc. 51)	Proposition 3.2, Theorems 5.2 and 5.3, Lemmas 6.4, 9.2 and 10.4, Theorem 10.11, Example 12.3	Match. Settles follow-up 3
Green–Tao–Ziegler	Published (Annals 176)	Example 6.11, the sets on p. 1272, Definitions 9.6 and 9.11, Lemmas 9.12 and 10.10, Proposition 8.3, the cited items of Appendices A, B and E	Labels and content are as the manuscript says, including both misprints in Example 6.11
Tao–Ziegler	Published (Forum Math. Pi 6)	Equations (32)–(34), Theorems 9, 11, 13 and 15, Corollary 3, Proposition 6	Match. Numbering as cited
Kolyvagin, ICM 1990	Max Planck preprint of the same text	Theorems 1–5	Match

Source	Version seen	What was compared	Outcome
Kato, Astérisque 295	Numdam scan	Corollary 14.3, §9.2, §9.4, displays (14.10.2)–(14.10.3)	Corollary 14.3, §9.2 and §9.4 match. The displays are present with the stated content, but their exponents and Frobenius convention are not legible in the scan's text layer
Dokchitser–Dokchitser	Published (Annals 172)	Theorem 1.4 (p. 568), Theorem 4.19 (p. 593)	Match
Poonen–Stoll	Published (Annals 150)	Corollaries 4 and 7, Theorem 8	Match
Koenigsmann	Published (Annals 183)	Definition 4, Proposition 6	Match
Voight, Quaternion algebras	Published, open access	Proposition 14.6.7	Match
Calegari, ICM 2022	Published	The modularity statement on p. 624	Match
Kopp–Lagarias	Published	Theorems 1.1, 1.3, 6.5 and 7.2, Remark 7.3	Match. Used only for the maximal order $\mathbb{Z}[i]$
Bienvenu	arXiv	Theorem 1.1	Match
Williams	Author's scan	Theorem 1 (p. 356)	Located. The formula is not legible in the text layer; it is Mertens' theorem in progressions
Deligne–Serre	Numdam scan	§§1.6–1.9, §3.1, Lemma 3.2, Theorem 4.1, §8	Match
Silverberg–Zarhin	Published	Theorems 3.3–3.5	Match
Scholze	arXiv	Corollary 1.8	Match. Theorems 9.1 and 9.3 not seen

Source	Version seen	What was compared	Outcome
Stacks Project	Site	Tags 0E83 and 0BW0	Match
Tointon	Author's copy of the book	Definition 5.4.3, Theorem 5.4.5, last paragraph of §5.4	Match
Gaudron–Rémond, Comment. Math. Helv.	Publisher's page	Bibliographic data for follow-up 2	Confirmed

Comparisons that still rest on a preprint. Pan (the article is in J. Amer. Math. Soc. 2022), Yuan (the published article is in Chinese), Rémond 2022, Browning–Loughran, Browning–Heath-Brown, Bienvenu, Scholze, and Kolyvagin as noted above. The publishers' copies were behind a paywall or needed an approval that timed out. I saw no sign that any published text differs. For the two Browning papers the manuscript's bibliography records that the project compared the published pages itself.

Not reached.

- Kato 1999 and Ribet 1990. See the next section.
- Mitsui 1956, Lemmas 14 and 15. The host refuses automated access.
- Colliot-Thélène–Salberger 1989, Proposition 1.3, and Colliot-Thélène–Sansuc–Swinnerton-Dyer 1987, Remark 2.3.1.
- Silverman's book. Every formal-group fact taken from it is re-derived above; the remaining citations are standard statements.
- The classical references behind the Shimura-curve constructions in the height chain: Faltings 1986, Shimura 1973, Igusa 1967, Buzzard 1997, Milne's notes on Shimura varieties, Deligne 1971 and 1979, Lin–Yang, Mascot, and the formula for the height of an elliptic curve with complex multiplication. The steps that use them stay *followed*, as in the third pass.

Kato 1999 and Ribet 1990

Still not read. For Ribet, both posted copies are image scans without a text layer, and the Göttingen viewer returned no page text. Reading the images needed a screenshot of your Chrome window, and every capture timed out, in this pass as in the last. I have closed the

tab I opened. For Kato, the publisher's page still returns a bot-protection page and I did not go around it. The repository holds only the project's own summary of the paper, which is not the source.

What depends on them is unchanged from the second pass: the repaired application at p = 2 in the restricted-converse argument rests on a reading of Kato's sections 0.1 and 0.6–0.8, and one step at lines 15105–15116 rests on Ribet's Propositions 3.3 and 3.6 and Remarks 3.5(a) and (b).

To close this, attach the two PDFs to the conversation. For Kato I need pages 314–318; for Ribet, pages 440–443.

Findings for the fourth pass

No error found. Every inference examined in this pass holds, and every number recomputed agrees with the text.

Changes to earlier findings

- The Bogolyubov citation at line 25988 is correct. That point is withdrawn.
- The fixed twenty-two-prime proposition (lines 2071–2530), which no earlier pass had audited, is verified.
- The majorant lemma in the primes chain moves from followed to verified, given the published estimates of Appendix D. The chain now has thirteen steps verified and three followed.
- The Mordell–Weil generator, the Kolyvagin step and the formal-group inputs move from not audited to verified.

Points for the authors

None of these affects a result.

- **Lines 26815 and 26883.** Insert the normalization paragraph and the added sentence given under follow-up 1.
- **Lines 30633 and 30646.** Cite the Commentarii paper, with the entry given under follow-up 2.
- **Lines 20063–20073.** Say that this adapts the proofs of Tao and Ziegler, or replace the adaptation by the reduction to their published statements sketched under follow-up 4, once it is written out.

What the results now rest on

For the 314-variable theorem and the 222-variable family, every part of the manuscript examined across the four passes holds. What they are conditional on:

1. **Published theorems used as stated.** Modularity (Breuil–Conrad–Diamond–Taylor, and Pan at the prime five), Kolyvagin's theorem, Yuan's isometry, the isogeny theorems of Gaudron and Rémond, the Green–Tao theorems, the Tao–Ziegler theorems and Mitsui's prime ideal theorem. I compared the statements with the sources, except Mitsui's and the classical references listed as not reached. I did not audit their proofs.
2. **The internal inverse theorem.** It has had one complete reading, mine, and needs the one inserted paragraph. The statement is also in the literature.
3. **The adaptation of the Tao–Ziegler proofs** to the modulus with a fixed extra factor. The changes are correct in outline; the proofs were not re-read with the changed modulus.
4. **Kato 1999 and Ribet 1990**, which I have not read.

What I would do next

- Attach Kato 1999 (pp. 314–318) and Ribet 1990 (pp. 440–443), and I will finish those two checks.
- Make the three edits listed under points for the authors.
- If someone has library access, the classical references in the not-reached list are the remaining source comparisons.